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101 Must-Know Challenging Maths Word Problems

Book 4

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Name: _____

Class: _____

Preface

101 Must-Know Challenging Maths Word Problems Book 4 presents word problems that test on important concepts so students can learn to apply general mathematical problem-solving strategies and heuristics confidently.

What's in this book?

This book comprises word problems often encountered by students in their tests and examinations. The questions are categorized into respective topics in accordance with the current **Primary Mathematics Syllabus**.

Solutions

Detailed step-by-step workings are included in the answer key for every question to show how a problem is solved. Diagrams and mathematical models are provided in most solutions to aid students in understanding the problem-solving processes.

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Question 1

On the first day of a school camp, there were 396 children. 27 boys left the camp on the second day. If there were twice as many boys as girls who remained at the camp, how many boys were there on the first day of the camp?

Answer: _____

Question 2

On Monday, Farmer Tom collected three times as many eggs as Farmer Jack. On Tuesday, Farmer Tom sold 160 eggs. Farmer Jack then had twice as many eggs as Farmer Tom. How many eggs did Farmer Jack have?

Answer: _____

Question 3

When George is 15 years old, his sister is 8 years old and his father is 45 years old. How old will George's sister be when George is half his father's age?

Answer: _____

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Question 4

Linda and Serene had an equal number of balloons. After Linda used 16 balloons, Serene had 3 times as many balloons as Linda. How many balloons did Serene have?

Answer: _____

Question 5

Jennifer had a party. She wanted to give an equal number of balloons to her six friends. If one of her friends did not turn up for the party, the rest would be able to receive another 2 balloons each. How many balloons did Jennifer have?

Answer: _____

Question 6

Anna had three times as many beads as Zoe. They had 480 beads altogether. Mrs Taylor gave an equal number of beads to each of them. Zoe then had half the number of beads that Anna had. How many beads did Mrs Taylor give them?

Answer: _____

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Question 7

Helen had to arrange some chairs in a fixed number of rows. She estimated that there were more than 30 but fewer than 70 chairs. If she put 8 chairs in one row, she would be 7 chairs short. If she put 7 chairs in one row, she would have 2 chairs left. How many chairs were there?

Answer: _____

Question 8

Zack and Mandy shared 120 sweets. After Zack gave Mandy 10 sweets, he had twice as many sweets as Mandy. How many sweets did each of them have at first?

Answers: Mandy: _____

Zack: _____

Question 9

A basket contained 4 times as many blue balls as red balls. After 21 blue balls were taken away, there were twice as many red balls as blue balls. How many balls were there in the basket at first?

Answer: _____

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Question 10

Nancy had $\frac{1}{3}$ as much money as Ben. Lucy had \$20 more than Ben. They had \$132 altogether. How much did each of them have?

Answers: Nancy: _____

Ben: _____

Lucy: _____

Question 11

There were three parcels, A, B and C. The mass of Parcel B was $\frac{3}{4}$ kg less than the mass of Parcel A. The mass of Parcel C was $\frac{2}{3}$ kg more than the mass of Parcel B. If the mass of Parcel A was $4\frac{1}{2}$ kg, what was the mass of Parcel C?

Answer: _____

Question 12

Mr Rice spent $\frac{1}{5}$ of his salary on food and $\frac{2}{3}$ on clothing. Half of the remaining amount was given to his wife and he saved the rest. If he had saved \$520, what was Mr Rice's salary?

Answer: _____

House Number Problem.
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Question 13

Sam had some marbles. He gave $\frac{1}{7}$ of them to his brother. $\frac{2}{3}$ of the remaining marbles were given to his cousin. The rest were kept in a box. If 48 marbles were kept in the box, how many more marbles did Sam give to his cousin than to his brother?

Answer: _____

Question 14

Mrs Anderson bought 60 eggs. She used $\frac{2}{5}$ of them to bake cakes. She used some eggs to bake cookies. She had 12 eggs left. How many eggs did she use for baking cookies?

Answer: _____

Question 15

$\frac{2}{7}$ of the members in a country club are men. 1050 women in the country club are below 50 years old. If there are 960 more women than men in the country club, how many women are above 50 years old?

Answer: _____

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Question 16

Stephanie had a total of 80 pieces of $2 and $10 notes. After spending $\frac{2}{5}$ of her $2 notes and 8 pieces of her $10 notes, she had an equal number of $2 and $10 notes left. How much money did Stephanie have at first?

Answer: _____

Question 17

$\frac{3}{5}$ of the animals on a farm are ducks and the rest are chickens. $\frac{1}{5}$ of the chickens are brown and the rest are white. If there are 24 white chickens, how many more ducks than chickens are there on the farm?

Answer: _____

Question 18

Susan earns $2450 a month. She spends $\frac{2}{5}$ of it and saves the rest. Andy earns $800 less than Susan. He spends $180 more than Susan each month and saves the rest. How much more can Susan save than Andy in half a year?

Answer: _____

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Question 19

On Monday, a shopkeeper sold $12\frac{1}{2}$ kg of flour. He sold $\frac{5}{6}$ kg more flour on Tuesday than on Monday. On Wednesday, he sold $1\frac{1}{3}$ kg less flour than on Tuesday. He then had $\frac{1}{3}$ kg of flour left in his shop. How much flour did he have at first?

Answer: _____

Question 20

$\frac{2}{7}$ of the toy cars in a box are silver. $\frac{1}{10}$ of the remainder are yellow and the rest are black. If there are 28 silver toy cars, how many black toy cars are there in the box?

Answer: _____

Question 21

Andrew and Ryan shared some chocolates. Andrew gave away $\frac{1}{4}$ of his chocolates. $\frac{1}{4}$ of the given chocolates were taken by Ryan. Ryan then had the same number of chocolates as Andrew. If Andrew gave away 12 chocolates, how many chocolates did each of them have at first?

Answers: Andrew: _____

Ryan: _____

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Question 22

Boxes A and B contained 112 pencils. When $\frac{1}{5}$ of the pencils in box A were transferred to box B, both boxes contained the same number of pencils. How many more pencils were there in box A than in box B at first?

Answer: _____

Question 23

Andrea spent $\frac{1}{5}$ of her salary on food and $\frac{1}{10}$ on transport. She gave the remaining amount to her three brothers equally. One of her brothers spent all his money on a radio and a watch that cost $86 and $145 respectively. What was Andrea's salary?

Answer: _____

Question 24

Ann and James had some stickers. After Ann gave James $\frac{1}{4}$ of her stickers, James had twice as many stickers as Ann. If they had a total of 117 stickers, how many stickers did each of them have in the beginning?

Answers: James: _____

Ann: _____

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Question 25

Mrs Brown gave $\frac{4}{7}$ of her money to her son. He spent $\frac{2}{5}$ of the money on a pair of shoes and the rest on two books that cost $18 each. How much did Mrs Brown have at first?

Answer: _____

Question 26

Ray sold $\frac{2}{3}$ of the charity tickets that Cindy sold. Cindy sold $\frac{3}{4}$ of the charity tickets that Gina sold. Each charity ticket cost $2. The children collected a total of $648. How many charity tickets did Gina sell?

Answer: _____

Question 27

There were $\frac{3}{4}$ as many sweets in box A as in box B. $\frac{1}{2}$ of the sweets in box A and $\frac{3}{8}$ of the sweets in box B were removed. There were then 152 sweets left in both boxes altogether. How many sweets were there in each box at first?

Answers: Box A: _____

Box B: _____

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Question 28

After their shopping trip, Mike spent $\frac{3}{5}$ of his money, Evon spent $\frac{1}{2}$ of hers and Sue spent $50. They then had the same amount of money left. If the total amount of money left was $54, what was the total amount of money Mike and Sue had at first?

Answer: _____

Question 29

A baker sold twice as many buns on Sunday than on Saturday. On Monday, he sold $\frac{1}{4}$ of the buns he sold on Sunday. If he sold a total of 1260 buns during the three days, how many buns did he sell on Sunday?

Answer: _____

Question 30

Joan and Alan had some money. $\frac{2}{3}$ of Joan's money was $\frac{1}{2}$ of what Alan had. If Alan spent all his money on a book that cost $45 and a wallet that cost $98, how much did Joan and Alan have altogether?

Answer: _____

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Question 31

Boxes A and B contained 150 marbles altogether. When $\frac{1}{3}$ of the marbles in box A were transferred to box B and 12 marbles were taken out from box B, there were twice as many marbles in box B than in box A. How many marbles were there in box B in the beginning?

Answer: _____

Question 32

During a sale, Betty bought a watch at $\frac{3}{5}$ its normal price. She then had $220 left. If she had bought the watch at its normal price, she would be $10 short. What was the normal price of the watch?

Answer: _____

Question 33

Mr Smith sold $\frac{1}{4}$ of his eggs on Monday and $\frac{1}{4}$ of the remainder on Tuesday. The rest were sold on Wednesday. If he sold 144 eggs on Wednesday, how many eggs did he have in the beginning?

Answer: _____

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Question 34

Sam could buy 4 magazines and 3 books with all his money. He decided to buy 2 magazines and a book. He then had $84 left. If a magazine cost $\frac{1}{3}$ the price of a book, how much did Sam have at first?

Answer: _____

Question 35

A farmer sold some eggs in the morning. He then packed the remaining eggs into 15 cartons with 25 eggs in each carton. If the number of eggs he sold was $\frac{3}{5}$ of the number of eggs he packed, how many eggs did the farmer have at first?

Answer: _____

Question 36

There were some candles in a box. $\frac{1}{3}$ of the candles were green. There were twice as many blue candles as red candles and $\frac{1}{4}$ as many yellow candles as red candles. If there were 24 red candles, how many candles were there in the box altogether?

Answer: _____

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Question 37

There were twice as many oranges as apples in a box and three times as many pears as apples. When $\frac{1}{4}$ of the oranges were eaten, there were 21 oranges left in the box. How many pieces of fruit were there in the box at first?

Answer: _____

Question 38

Elaine had some sweets. When she ate $\frac{2}{7}$ of them and gave 36 sweets to her brother, she had $\frac{2}{7}$ of the sweets left. How many sweets did she have at first?

Answer: _____

Question 39

Jug A contained $3\frac{5}{12}$ *l* of water. Jug A contained $1\frac{2}{3}$ *l* less water than jug B. Jug C contained $\frac{5}{6}$ *l* less water than jug B. The water in the three jugs was poured into a container. How much water was there in the container in the end?

Answer: _____

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Question 40

Mrs Crawford baked some pies. She gave $\frac{1}{3}$ of them to her neighbour. She then gave $\frac{5}{12}$ of the remainder to a children's home. The rest of the pies were packed into 25 boxes with 14 pies in each box. How many pies did Mrs Crawford bake altogether?

Answer: _____

Question 41

Sandy spent $\frac{3}{5}$ of her money in a toy shop on a teddy bear for $38 and a toy train for $46. The remaining money was then spent on 5 similar books. How much did each book cost?

Answer: _____

Question 42

Jeff sold 120 oranges on Friday. He sold 100 fewer oranges on Friday than on Saturday. The number of oranges he sold on Sunday was three times the number he sold on Friday. If he had $\frac{2}{7}$ of the oranges left, how many oranges did he have at first?

Answer: _____

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Question 43

There were 120 more eggs in box A than in box B. There were $\frac{2}{3}$ as many eggs in box C than in box B. When half of the eggs in box B were sold, there were 222 eggs left in box B. How many eggs were there altogether?

Answer: _____

Question 44

Alfred had 72 mugs. He sold some of them for $144. He then had $\frac{7}{8}$ of the mugs left. How much would he get if he had sold all the mugs?

Answer: _____

Question 45

There were some chicken wings and fishballs in a pack. After $\frac{2}{7}$ of the chicken wings were taken out from the pack, there were $\frac{3}{10}$ as many chicken wings as fishballs in the pack. If there were 50 fishballs, how many chicken wings were there in the pack at first?

Answer: _____

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Question 46

There were half as many yellow T-shirts as blue T-shirts in a box and 15 more green T-shirts than blue T-shirts. There were 27 yellow T-shirts. How many T-shirts were left in the box when $\frac{4}{5}$ of the T-shirts were taken out?

Answer: _____

Question 47

There were some pears in boxes A and B. When $\frac{1}{3}$ of the pears were taken out from box A and $\frac{5}{9}$ of the pears were taken out from box B, there was an equal number of pears in both boxes. If 84 pears were taken out, how many pears were there in both boxes at first?

Answer: _____

Question 48

Agnes had a total of 135 balloons. There were three times as many red balloons as green balloons, twice as many blue balloons as green balloons and some purple balloons. When 12 green balloons burst, there were $\frac{1}{3}$ as many green as purple balloons left. Find the total number of purple and green balloons Agnes had at first.

Answer: _____

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Question 49

Michael paid $14 for 3 similar mangoes and a durian. The durian cost $2 more than each mango. What was the total cost of 5 mangoes and 4 durians?

Answer: _____

Question 50

A blouse costs half as much as a shirt. If the total cost of 3 similar blouses and a shirt is $200, what is the cost of 7 blouses and 4 shirts?

Answer: _____

Question 51

Colin paid $12 for 4 buns and 4 cakes. 3 buns cost as much as 2 cakes. What was the total cost of 6 buns and 9 cakes?

Answer: _____

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Question 52

The total cost of a radio, a clock and a blender is $215. The radio costs $35 more than the blender. The blender costs $45 more than the clock. What is the cost of the radio?

Answer: _____

Question 53

Betty had twice as much money as Johnson. After Betty spent $350, she had $25 less than Johnson. Find the total amount of money they had at first.

Answer: _____

Question 54

Connie and Alice had an equal amount of money at first. Connie gave Alice $126 and Alice had 4 times as much money as Connie in the end. How much money did Connie have at first?

Answer: _____

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Question 55

Terence and Winnie had a total of 27 coins. When Terence gave Winnie $\frac{1}{4}$ of his coins, she had twice as many coins as him. Winnie then found out that she had twice as many twenty-cent coins as ten-cent coins. How much money did she have in the end?

Answer: _____

Question 56

Hazel had enough money to buy 25 chocolate cookies. Each chocolate cookie cost as much as 2 butter cookies. Hazel borrowed another $6 from her mother to buy 20 chocolate cookies and 20 butter cookies. How much money did Hazel have at first?

Answer: _____

Question 57

A book costs three times as much as a pen. 5 similar pens cost as much as a photo frame. If 3 similar photo frames cost $74.25, what is the total cost of a book, a pen and a photo frame?

Answer: _____

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Question 58

For every T-shirt that Joel sells, he will receive $12. He will receive another $5 for every set of 10 T-shirts he sells. If he sells 100 T-shirts, he will receive an additional $50 bonus. How many T-shirts does Joel have to sell to receive $1372 in total?

Answer: _____

Question 59

4 boxes of coloured pencils cost as much as 3 boxes of paint. Natalie had just enough money to buy 7 boxes of paint and 2 boxes of coloured pencils. If a box of paint cost $5.40, how much would Natalie have left if she only bought 5 boxes of paint?

Answer: _____

Question 60

Jeffrey had 4 times as many twenty-cent coins as ten-cent coins and $\frac{1}{3}$ as many fifty-cent coins as ten-cent coins. If he had 2 fifty-cent coins, how much money would he have left if he spent $2.75 on a pair of socks?

Answer: _____

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Question 61

Michael had some flowers. He sold $\frac{2}{3}$ of them at 90¢ each and the rest at 50¢ each. Fanny sold the same number of flowers at 80¢ each, collecting a total of $7.20. How much did Michael collect?

Answer: _____

Question 62

When Nelly bought 5 pencils, she spent the same amount as Benjamin who bought 3 pens. If Benjamin spent $\frac{2}{7}$ of his money buying the pens and he had $45 left, what was the total cost of a pencil and a pen?

Answer: _____

Question 63

Ron earns $12 an hour on weekdays and twice as much on weekends. Ron works 8 hours a day on weekdays and 5 hours a day on Saturdays and Sundays. How much will he earn in 2 weeks if he works every day?

Answer: _____

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Question 64

The cost of a magazine and 2 books is $34. The cost of 3 magazines and 4 books is $76. If Tasha buys 13 magazines and 9 books, how much money does she spend?

Answer: _____

Question 65

An apple and two oranges cost $2.30. Two apples and an orange cost $2.50. What is the cost of 4 apples and 6 oranges?

Answer: _____

Question 66

Mrs Bellamont sold twice as many butter cookies as chocolate chip cookies and twice as many chocolate chip cookies as coconut cookies. Each box of 4 cookies was sold for $4.50. 80 chocolate chip cookies were sold. How much would Mrs Bellamont collect from the sale of all the cookies?

Answer: _____

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Question 67

The total mass of Gilbert and Sean was 124 kg. If Sean lost $\frac{1}{4}$ of his mass, he would be 45 kg. How much heavier was Gibert compared to Sean?

Answer: _____

Question 68

4 parcels, A, B, C and D have a total mass of 100.75 kg. Parcel A weighs 12.45 kg more than parcel B. Parcel B weighs 8.4 kg less than parcel C. Parcel C weighs 24.2 kg. How much heavier is parcel D than parcel C?

Answer: _____

Question 69

Box A was 3.45 kg heavier than box B. When 1.2 kg of items in box A was removed, box A became four times as heavy as box B. What was the mass of box A in the beginning?

Answer: _____

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Question 70

The mass of a box is 45 kg when it is full. When half of the items in the box are taken out, the mass of the remaining items and the box is 33 kg. What is the mass of 6 empty boxes?

Answer: _____

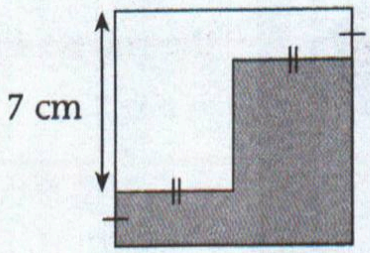
Question 71

9 small squares are used to form a large square. If each side of a small square is 7 cm, what is the area of the large square?

Answer: _____

Question 72

The area of a square is 81 cm^2. Find the perimeter of the unshaded portion.



Answer: _____

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Question 73

The perimeter of a rectangle is 288 cm. The length is three times the breadth. Find the area of the rectangle.

Answer: _____

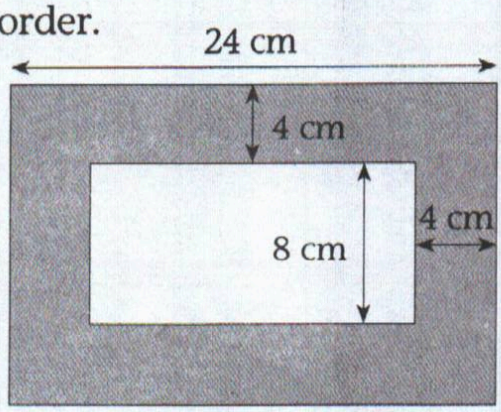
Question 74

A wire was cut and $\frac{2}{3}$ of it was bent to form a square. The remaining length of the wire was bent to form a rectangle. The area of the square was 36 cm^2. What was the length of the wire at first?

Answer: _____

Question 75

Jenny draws a picture on a piece of paper with a border of 4 cm around the picture. The length of the paper is 24 cm. If the breadth of the picture is 8 cm, find the area of the border.



Answer: _____

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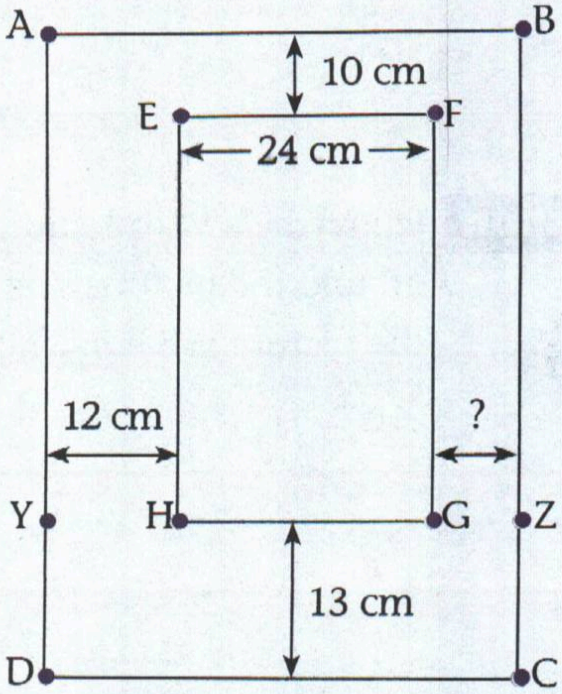
Question 76

The perimeter of a square is half the area of a rectangle. A wire is bent to form 3 such squares and 2 such rectangles. 20 cm of wire is left. If the length of the rectangle is 15 cm and its breadth is 12 cm, find the length of the wire.

Answer: _____

Question 77

The perimeter of rectangle EFGH is 180 cm. YH is twice GZ. If EF is 24 cm and YH is 12 cm, what is the area of rectangle ABCD?



Answer: _____

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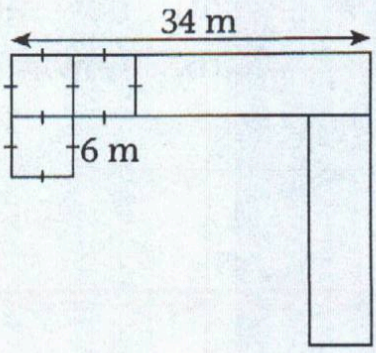
Question 78

Mr Cox wants to put a fence around his garden. His garden is in the shape of two similar rectangles and three similar squares. If it costs $23 to fence two metres of his garden, how much will Mr Cox have to pay to fence the entire garden?



Answer: _____

Question 79

Mr Richard is five times as old as his son now. In 8 years' time, their total ages will be 58 years.

(a) What is his son's present age?

(b) How old will Mr Richard be in 8 years' time?

Answers: (a) _____

(b) _____

Question 80

Jane and Mark had $240 altogether. When Jane gave Mark half of her money, Mark had four times as much money as Jane.

(a) How much did Jane have at first?

(b) How much did Mark have at first?

Answers: (a) _____

(b) _____

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Question 81

Mrs Fleming spent $\frac{1}{3}$ of her money on a handbag and $\frac{3}{4}$ of the remainder on jewellery. She then saved the rest. If she spent \$450 on a bracelet and \$90 on a pair of earrings,

(a) how much money did she save?

(b) how much money did she have at first?

Answers: (a) _____

(b) _____

Question 82

Sean had some money. He used $\frac{2}{3}$ of it to buy a watch and spent $\frac{3}{4}$ of the remainder on a wallet. If the watch cost \$65 more than the wallet,

(a) how much money had Sean left?

(b) how much money did Sean have at first?

Answers: (a) _____

(b) _____

Question 83

$\frac{1}{3}$ of the books on a shelf are English books. $\frac{1}{2}$ of the remainder are Chinese books and the rest are Malay books. $\frac{1}{4}$ of the Chinese books are torn. If 54 Chinese books are not torn,

(a) how many Malay books are there?

(b) how many books are there altogether?

Answers: (a) _____

(b) _____

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Question 84

There were 250 students in a room. $\frac{2}{5}$ of the girls and $\frac{1}{10}$ of the boys wore spectacles. If the same number of boys and girls did not wear spectacles,

(a) how many boys wore spectacles?

(b) how many girls wore spectacles?

Answers: (a) _____

(b) _____

Question 85

At a football match, there were three times as many men as women and twice as many men as boys. $\frac{1}{3}$ of the children were girls. If there were 120 girls,

(a) how many boys were there?

(b) how many adults were there?

Answers: (a) _____

(b) _____

Question 86

$\frac{2}{3}$ of the people at a fair wore hats. $\frac{1}{6}$ of those who wore hats were men and the rest were women. There were 124 more women than men who wore hats at the fair.

(a) How many people did not wear hats?

(b) How many people were there at the fair?

Answers: (a) _____

(b) _____

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Question 87

Two numbers are written on a piece of paper. $\frac{1}{5}$ of the first number is 12 more than $\frac{1}{2}$ of the second number. The sum of the two numbers is 221.

(a) What is the first number?

(b) What is the second number?

Answers: (a) _____

(b) _____

Question 88

Gracia had some pens. $\frac{2}{5}$ of the pens were blue and $\frac{1}{6}$ of the remainder were black. The rest were red and green. There were 12 more red pens than blue pens. If there were 24 green pens,

(a) how many red and blue pens did Gracia have?

(b) how many pens did Gracia have altogether?

Answers: (a) _____

(b) _____

Question 89

The total number of oranges in cartons A and B was 324. When $\frac{3}{7}$ of the oranges in carton A were sold, there were twice as many oranges in carton A than in carton B.

(a) How many oranges were there in carton A at first?

(b) How many oranges were there in carton B at first?

Answers: (a) _____

(b) _____

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Question 90

On Monday, Mr Hall sold 246 lemons and Mr Clement sold half as many lemons as Mr Hall. On Tuesday, Mr Hall sold 83 lemons and Mr Clement sold some lemons. The total number of lemons sold by Mr Clement over the two days was $\frac{4}{7}$ the number that Mr Hall sold.

(a) How many lemons did Mr Clement sell on Tuesday?

(b) How many lemons did both men sell altogether on both days?

Answers: (a) _____

(b) _____

Question 91

Cleo and Diana were given the same amount of money each day. Cleo spent $\frac{4}{7}$ of her money and Diana spent $\frac{2}{3}$ of her money every day. They then saved the rest of their money. After a week, Cleo saved $56 more than Diana.

(a) How much money did Cleo spend a day?

(b) How much money did Diana spend a day?

Answers: (a) _____

(b) _____

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Question 92

A box of apples costs $\frac{2}{7}$ the price of a box of oranges. There are $\frac{1}{3}$ as many apples as oranges in one box. If there are 90 oranges in one box and an orange costs 70¢,

(a) how much does a box of apples cost?

(b) how many apples are there in one box?

Answers: (a) _____

(b) _____

Question 93

Katherine and Leslie had a total of 140 sweets. When Leslie gave Katherine $\frac{1}{4}$ of his sweets and Katherine ate 5 of her sweets, Katherine had twice as many sweets as Leslie.

(a) How many sweets did Leslie have at first?

(b) How many sweets did Katherine have at first?

Answers: (a) _____

(b) _____

More on Fractions.
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Question 94

There were some pears and apples in a box. If 4 pears were taken out from the box, there would be an equal number of pears and apples in the box. If 7 pears were taken out from the box, there would be $\frac{4}{5}$ as many pears as apples in the box.

(a) How many apples were there in the box at first?

(b) How many pears were there in the box at first?

Answers: (a) _____

(b) _____

Question 95

There were some spectators at a tennis match. $\frac{1}{3}$ of them belonged to the tennis club. $\frac{1}{6}$ of those who did not belong to the club were adults and the rest were children. If there were 120 more children than adults who did not belong to the club,

(a) how many people belonged to the tennis club?

(b) how many spectators were there at the tennis match?

Answers: (a) _____

(b) _____

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Question 96

Five apples cost as much as two papayas. Mrs Lynch could buy 18 papayas. If she had bought 12 papayas, she would have $30 left.

(a) How much did each papaya cost?

(b) How much would three apples and three papayas cost?

Answers: (a) _____

(b) _____

Question 97

Joey had some twenty-cent and ten-cent coins. When she used $\frac{1}{3}$ of them, she had $9.60 left. She used half as many twenty-cent coins as ten-cent coins.

(a) How many twenty-cent coins did she use?

(b) How many ten-cent coins did she use?

Answers: (a) _____

(b) _____

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Question 98

James spent $\frac{1}{5}$ of his money on a pair of shoes and $\frac{1}{10}$ on a book. He saved $\frac{2}{7}$ of the remaining money and the rest of the money was given to and shared equally among his 8 brothers. If each brother received $5,

(a) how much money did James have at first?

(b) how much money did James spend?

Answers: (a) _____

(b) _____

Question 99

Vivian and her brother had some money. After spending $\frac{2}{5}$ of his money, her brother had $78 left. Vivian spent $\frac{1}{3}$ of her money and had the same amount of money left as her brother.

(a) How much money did Vivian have at first?

(b) How much money did Vivian's brother have at first?

Answers: (a) _____

(b) _____

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Question 100

A piece of wire is cut into two equal pieces. One piece is bent to form two similar squares with sides 12 cm. The other piece is bent into three similar rectangles, each with a breadth of 6 cm.

(a) Find the length of the piece of wire at first.

(b) Find the area of each rectangle.

Answers: (a) _____

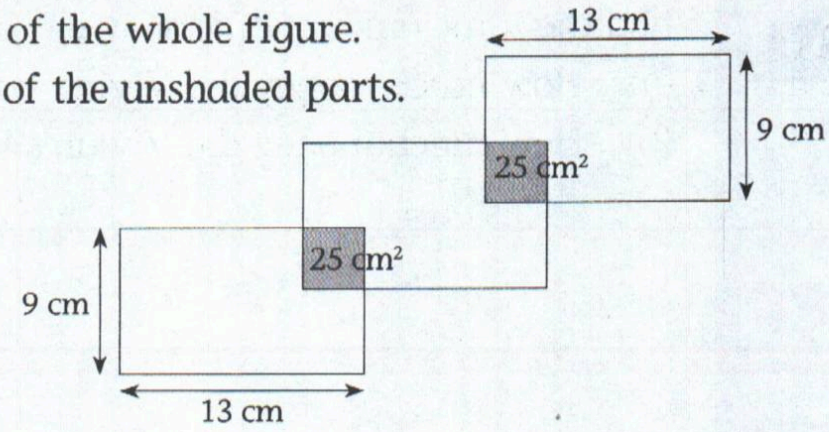
(b) _____

Question 101

The figure shows 3 identical rectangles. The shaded areas where the rectangles overlap form squares of 25 cm^2 each.

(a) Find the perimeter of the whole figure.

(b) Find the total area of the unshaded parts.



Answers: (a) _____ cm

(b) _____ cm^2

More on Area and Perimeter.
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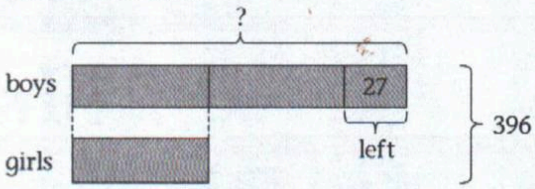
Solution to Question 1

Step 1 : Draw a model



Step 2 : Find how many children remained at the camp

396 – 27 = 369

Step 3 : Find the number of girls at the camp

3 units → 369

1 unit → 369 ÷ 3 = 123

Step 4 : Find the number of boys there were on the first day of camp

2 units → 2 × 123 = 246

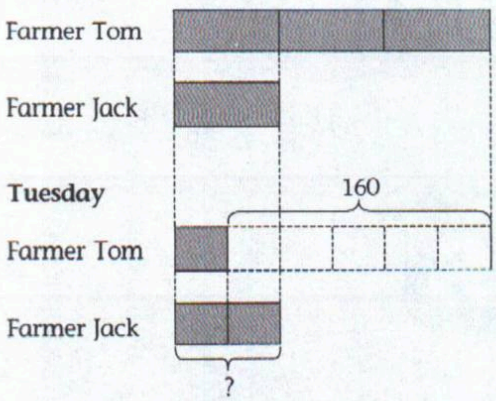
246 + 27 = 273

There were 273 boys on the first day of the camp.

Answer: 273 boys

Solution to Question 2

Step 1 : Draw a model



Step 2 : Find the number of eggs Farmer Tom had left

5 units → 160

1 unit → 160 ÷ 5 = 32

Step 3 : Find the number of eggs Farmer Jack had

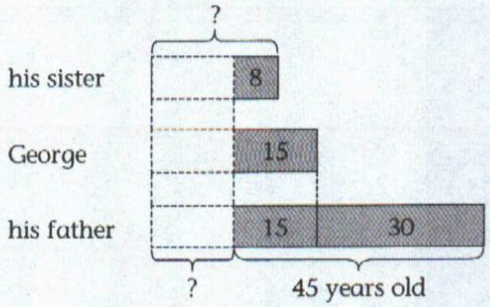
2 units → 2 × 32 = 64

Farmer Jack had 64 eggs.

Answer: 64 eggs

Solution to Question 3

Step 1 : Draw a model



Step 2 : Find the number of years for George to be half of his father's age

1 unit + 15 = 30

1 unit → 30 – 15 = 15

Step 3 : Find his sister's age in 15 years' time

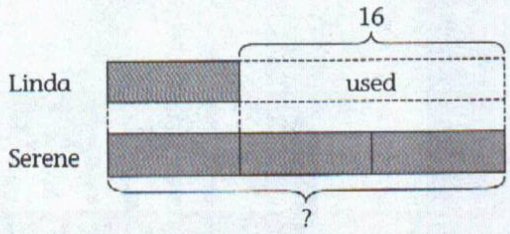
15 + 8 = 23

George's sister will be 23 years old when George is half his father's age.

Answer: 23 years old

Solution to Question 4

Step 1 : Draw a model



Step 2 : Find the number of balloons Linda had after using 16 balloons

2 units → 16

1 unit → 16 ÷ 2 = 8

Step 3 : Find the number of balloons Serene had

3 units → 3 × 8 = 24

Serene had 24 balloons.

Answer: 24 balloons

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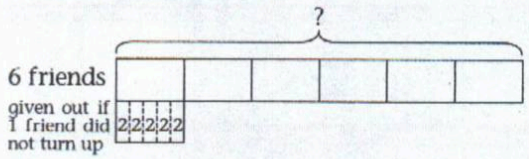
Solution to Question 5

Step 1 : Draw a model



Step 2 : Find the number of balloons that would be given out if 1 friend did not turn up

$5 \times 2 = 10$

Step 3 : Find the number of balloons that Jennifer had

$6 \times 10 = 60$

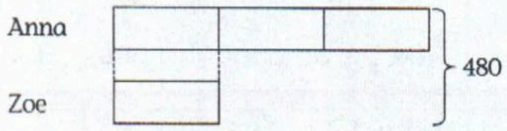
Jennifer had 60 balloons.

Answer: 60 balloons

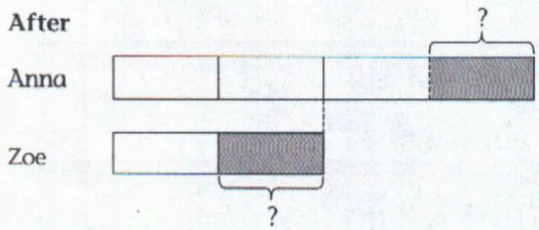
Solution to Question 6

Step 1 : Draw a model

Before



After



Step 2 : Find the number of beads Zoe had

4 units → 480

1 unit → 480 ÷ 4 = 120

Step 3 : Find the total number of beads that Mrs Taylor gave them

(*From the model, Anna has 2 units more than Zoe even after receiving the same number of beads from Mrs Taylor.)

2 units → 2 × 120 = 240

Mrs Taylor gave them 240 beads.

Answer: 240 beads

Solution to Question 7

Thinking skills:
Spatial Visualisation

Method 1

Step 1 : Use the 'Guess and Check' method to find the total number of chairs

8 chairs in 1 row

Number of rows	1	2	3	4	5	6	7	8	9
Number of chairs	8	16	24	32	40	48	56	64	72
Number of chairs short	7	7	7	7	7	7	7	7	7
Total number of chairs	1	9	17	25	33	41	49	57	(65)

(−) (−)

7 chairs in 1 row

Number of rows	1	2	3	4	5	6	7	8	9
Number of chairs	7	14	21	28	35	42	49	56	63
Number of chairs left	2	2	2	2	2	2	2	2	2
Total number of chairs	9	16	23	30	37	44	51	58	(65)

(+) (+)

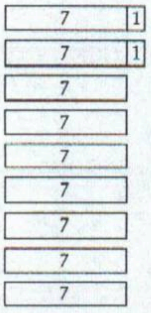
Number of rows : 9

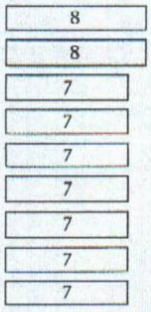
Total number of chairs : 65

Method 2

Step 1 : Draw diagrams

If she placed 7 chairs in one row, she would have 2 chairs left.





If she placed 8 chairs in one row, she would be 7 chairs short.

Step 2 : Find the total number of chairs (see diagram on the left)

$7 \times 9 = 63$

$63 + 2 = 65$

or Find the total number of chairs (see diagram on the right)

$8 \times 2 = 16$

$7 \times 7 = 49$

$16 + 49 = 65$

There were 65 chairs.

Answer: 65 chairs

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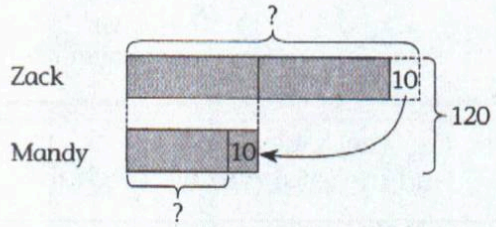
Solution to Question 8

Step 1 : Draw a model



Step 2 : Find the number of sweets Mandy had in the end

3 units → 120

1 unit → 120 ÷ 3 = 40

Step 3 : Find the number of sweets Mandy had at first

40 – 10 = 30

Step 4 : Find the number of sweets Zack had at first

2 units → 2 × 40 = 80

80 + 10 = 90

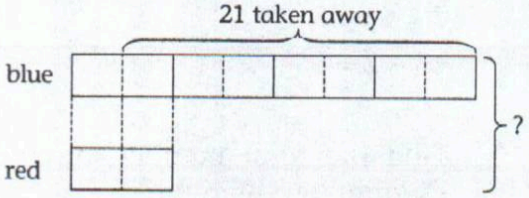
Mandy had 30 sweets and Zack had 90 sweets at first.

Answers: Mandy: 30 sweets

Zack: 90 sweets

Solution to Question 9

Step 1 : Draw a model



Step 2 : Find the total number of balls at first

7 units → 21

1 unit → 21 ÷ 7 = 3

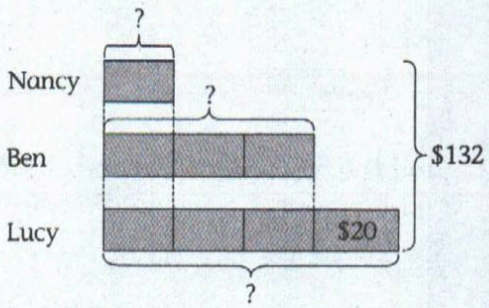
10 units → 10 × 3 = 30

There were 30 balls in the basket at first.

Answer: 30 balls

Solution to Question 10

Step 1 : Draw a model



Step 2 : Find the amount of money Nancy had

$132 – $20 = $112

7 units → $112

1 unit → $112 ÷ 7 = $16

Step 3 : Find the amount of money Ben had

3 units → 3 × $16 = $48

Step 4 : Find the amount of money Lucy had

$48 + $20 = $68

Nancy had $16.

Ben had $48.

Lucy had $68.

Answers: Nancy: $16

Ben: $48

Lucy: $68

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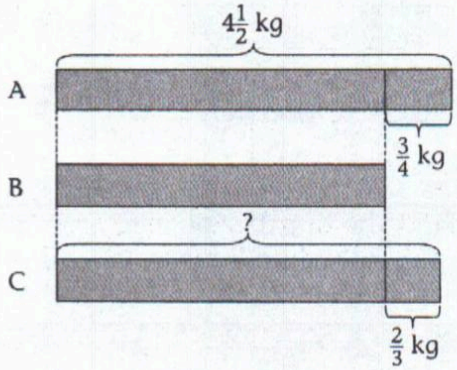
Solution to Question 11

Step 1 : Draw a model



Step 2 : Find the mass of parcel B

Remember to change to a common denominator before subtracting

$$4\tfrac{1}{2}\text{ kg} - \tfrac{3}{4}\text{ kg} = \tfrac{9^{\times 2}}{2_{\times 2}}\text{ kg} - \tfrac{3}{4}\text{ kg}$$
$$= \tfrac{18}{4}\text{ kg} - \tfrac{3}{4}\text{ kg}$$
$$= \tfrac{15}{4}\text{ kg}$$
$$= 3\tfrac{3}{4}\text{ kg}$$

$1 = \frac{4}{4}$

Step 3 : Find the mass of parcel C

Remember to change to a common denominator before adding

$$3\tfrac{3}{4}\text{ kg} + \tfrac{2}{3}\text{ kg} = \tfrac{15^{\times 3}}{4_{\times 3}}\text{ kg} + \tfrac{2^{\times 4}}{3_{\times 4}}\text{ kg}$$
$$= \tfrac{45}{12}\text{ kg} + \tfrac{8}{12}\text{ kg}$$
$$= \tfrac{53}{12}\text{ kg}$$
$$= 4\tfrac{5}{12}\text{ kg}$$

$1 = \frac{12}{12}$

The mass of Parcel C was $4\frac{5}{12}$ kg .

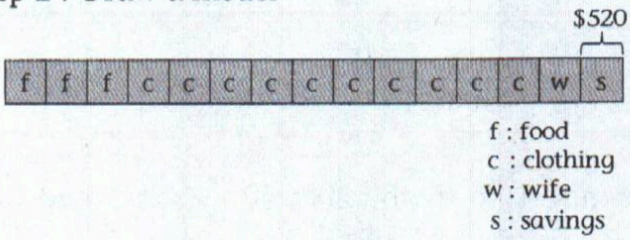
Answer: $4\frac{5}{12}$ kg

Solution to Question 12

Step 1 : Change $\frac{1}{5}$ and $\frac{2}{3}$ into equivalent fractions

$$\frac{1^{\times 3}}{5_{\times 3}} = \frac{3}{15}$$
$$\frac{2^{\times 5}}{3_{\times 5}} = \frac{10}{15}$$

Step 2 : Draw a model



Step 3 : Find Mr Rice's salary

1 unit → \$520
15 units → 15 × \$520
= \$7800

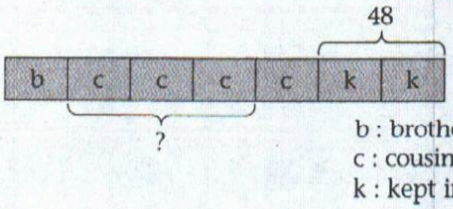
Mr Rice's salary was \$7800.

Answer: \$7800

Solution to Question 13

Step 1 : Draw a model

$\frac{2}{3} = \frac{4}{6}$



Step 2 : Find the number of marbles Sam gave to his brother

2 units → 48
1 unit → 48 ÷ 2 = 24

Step 3 : Find how many more marbles Sam gave to his cousin than to his brother

4 units – 1 unit = 3 units
3 units → 3 × 24 = 72

Sam gave 72 more marbles to his cousin than to his brother.

Answer: 72 marbles

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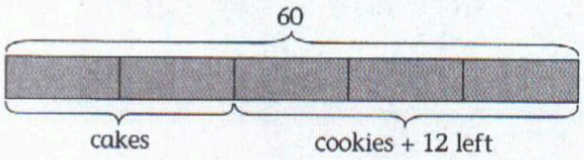
Solution to Question 14

Step 1 : Draw a model



Step 2 : Find the number of eggs she used to bake cakes

5 units → 60
1 unit → 60 ÷ 5 = 12
2 units → 2 × 12 = 24

Step 3 : Find the number of eggs she used for baking cookies

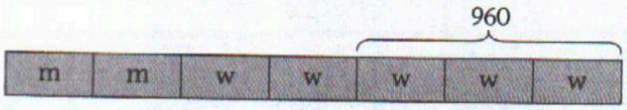
60 – 24 – 12 = 24

She used 24 eggs for baking cookies.

Answer: 24 eggs

Solution to Question 15

Step 1 : Draw a model



m : men
w : women

Step 2 : Find the total number of women in the country club

3 units → 960
1 unit → 960 ÷ 3 = 320
5 units → 5 × 320 = 1600

Step 3 : Find the number of women who are above 50 years old

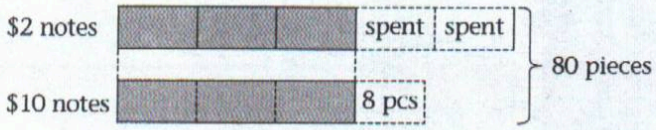
1600 – 1050 = 550

550 women are above 50 years old.

Answer: 550 women

Solution to Question 16

Step 1 : Draw a model



Step 2 : Find the number of pieces of notes left after spending 8 pieces of $10 notes

80 – 8 = 72

Step 3 : Find the number of pieces of $2 notes

8 units → 72
1 unit → 72 ÷ 8 = 9
5 units → 5 × 9 = 45

Step 4 : Find the number of $10 notes

3 units → 3 × 9 = 27
27 + 8 = 35

Step 5 : Find the amount of money she had at first

45 × $2 = $90
35 × $10 = $350
$90 + $350 = $440

Stephanie had $440 at first.

Answer: $440

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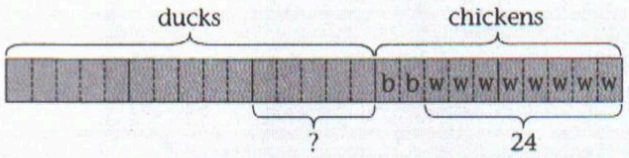
Solution to Question 17

Step 1 : Draw a model

$\frac{2}{5} = \frac{10}{25}$

$\frac{1}{5} = \frac{2}{10}$

*Cut 1 big unit into 5 smaller units



b : brown chickens
w : white chickens

Step 2 : Find how many more ducks than chickens there are

chickens → 10 units

ducks → 15 units

15 – 10 = 5 units more

8 units → 24

1 unit → 24 ÷ 8 = 3

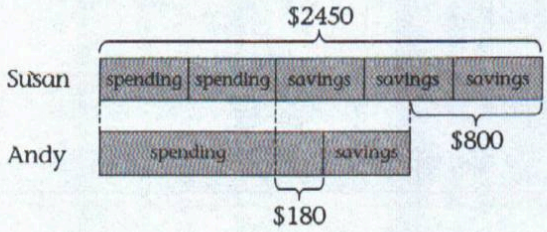
5 units → 5 × 3 = 15

There are 15 more ducks than chickens on the farm.

Answer: 15 more ducks

Solution to Question 18

Step 1 : Draw a model



Step 2 : Find the amount of money Susan saves in a month

5 units → $2450

1 unit → $2450 ÷ 5 = $490

3 units → 3 × $490 = $1470

Step 3 : Find the amount of money Andy saves in a month

$1470 – $800 – $180 = $490

Step 4 : Find how much more Susan can save than Andy in half a year

$1470 – $490 = $980

6 × $980 = $5880

Susan can save $5880 more than Andy in half a year.

Answer: $5880

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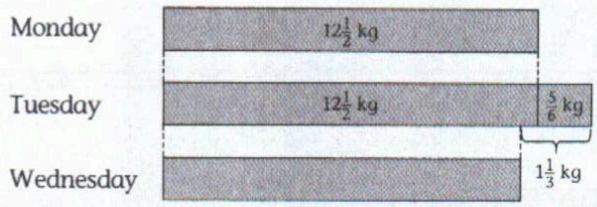
Solution to Question 19

Step 1 : Draw a model



Step 2 : Find the amount of flour sold on Tuesday

Remember to change to a common denominator before adding

$12\frac{1}{2}$ kg + $\frac{5}{6}$ kg = $\frac{25}{2}$ kg + $\frac{5}{6}$ kg

= $\frac{75}{6}$ kg + $\frac{5}{6}$ kg

= $\frac{80}{6}$ kg

= $13\frac{2}{6}$ kg

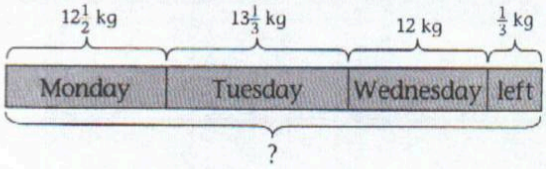
= $13\frac{1}{3}$ kg

$\frac{2}{6} = \frac{1}{3}$

Step 3 : Find the amount of flour sold on Wednesday

$13\frac{1}{3}$ kg – $1\frac{1}{3}$ kg = 12 kg

Step 4 : Find the total amount of flour he had at first



$12\frac{1}{2}$ kg + $13\frac{1}{3}$ kg + 12 kg + $\frac{1}{3}$ kg

= $12\frac{3}{6}$ kg + $13\frac{2}{6}$ kg + 12 kg + $\frac{2}{6}$ kg

= $37\frac{7}{6}$ kg

= $38\frac{1}{6}$ kg

$\frac{7}{6} = 1\frac{1}{6}$

He had $38\frac{1}{6}$ kg of flour at first.

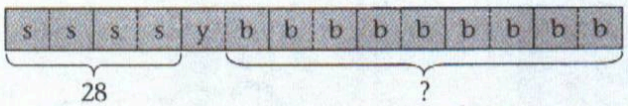
Answer: $38\frac{1}{6}$ kg of flour

Solution to Question 20

Step 1 : Draw a model

Remainder
$\frac{7}{7} - \frac{2}{7} = \frac{5}{7} = \frac{10}{14}$

$\frac{2}{7} = \frac{4}{14}$



28

?

s : silver toy cars
y : yellow toy cars
b : black toy cars

Step 2 : Find the number of black toy cars in the box

4 units → 28

1 unit → 28 ÷ 4 = 7

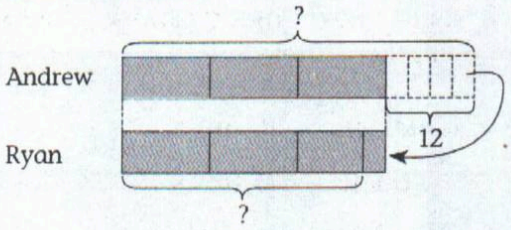
9 units → 9 × 7 = 63

There are 63 black toy cars in the box.

Answer: 63 black toy cars

Solution to Question 21

Step 1 : Draw a model



Step 2 : Find the number of chocolates Andrew had at first

1 unit → 12

4 units → 4 × 12 = 48

Step 3 : Find the number of chocolates taken by Ryan

$\frac{1}{\cancel{4}_{1}} \times \cancel{12}^{3} = 3$

Step 4 : Find the number of chocolates Ryan had at first

3 units → 3 × 12 = 36

36 – 3 = 33

Andrew had 48 chocolates and Ryan had 33 chocolates at first.

Answers: Andrew: 48 chocolates

Ryan: 33 chocolates

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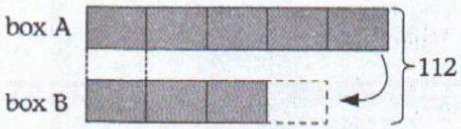
Solution to Question 22

Step 1 : Draw a model



Step 2 : Find the number of pencils transferred to box B

8 units → 112
1 unit → 112 ÷ 8 = 14

Step 3 : Find how many more pencils there were in box A than box B at first

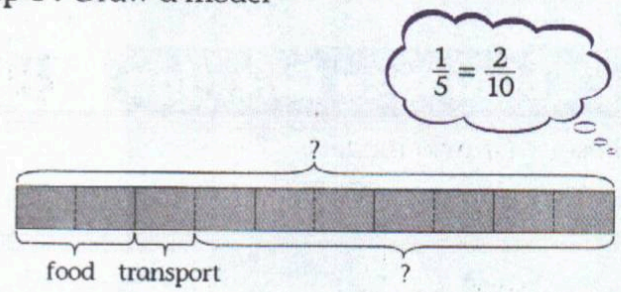
2 units → 2 × 14 = 28

There were 28 more pencils in box A than in box B at first.

Answer: 28 pencils

Solution to Question 23

Step 1 : Draw a model



Step 2 : Find the amount of money she gave to her three brothers

$86 + $145 = $231
3 × $231 = $693

Step 3 : Find Andrea's salary

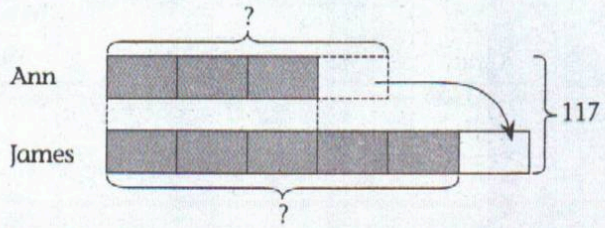
7 units → $693
1 unit → $693 ÷ 7
= $99
10 units → 10 × $99
= $990

Andrea's salary was $990.

Answer: $990

Solution to Question 24

Step 1 : Draw a model



Step 2 : Find the number of stickers given by Ann

9 units → 117
1 unit → 117 ÷ 9 = 13

Step 3 : Find the number of stickers for each of them

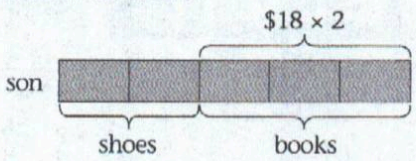
James → 5 × 13 = 65
Ann → 4 × 13 = 52

James had 65 stickers and Ann had 52 stickers in the beginning.

Answers: James: 65 stickers
Ann: 52 stickers

Solution to Question 25

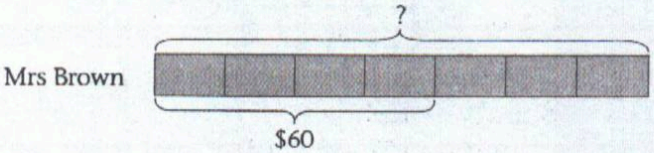
Step 1 : Draw a model



Step 2 : Find the amount of money her son received

3 units → 2 × $18
= $36
1 unit → $36 ÷ 3
= $12
5 units → 5 × $12
= $60

Step 3 : Draw another model



Step 4 : Find the amount of money Mrs Brown had at first

4 units → $60
1 unit → $60 ÷ 4 = $15
7 units → 7 × $15 = $105

Mrs Brown had $105 at first.

Answer: $105

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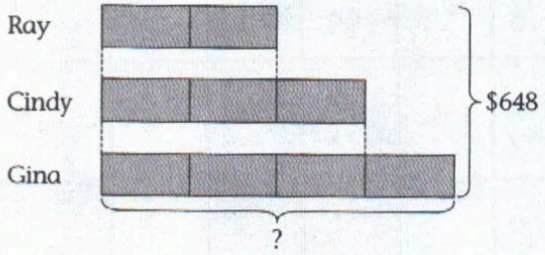
Solution to Question 26

Step 1 : Draw a model



Step 2 : Find the total number of charity tickets that three of them sold

$648 ÷ $2 = 324 tickets

Step 3 : Find the number of charity tickets that Gina sold

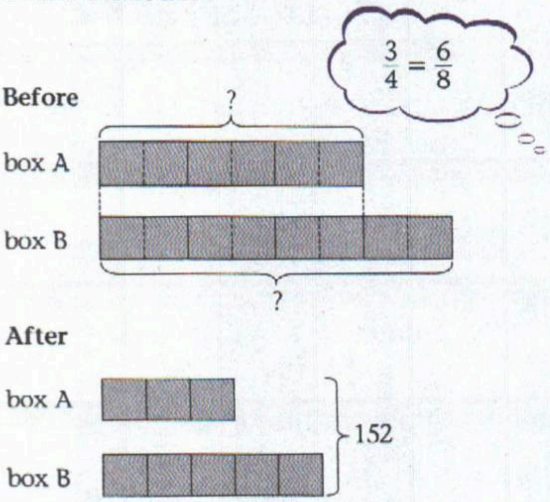
9 units → 324
1 unit → 324 ÷ 9
= 36
4 units → 4 × 36
= 144

Gina sold 144 charity tickets.

Answer: 144 charity tickets

Solution to Question 27

Step 1 : Draw a model



Step 2 : Find the number of sweets for 1 unit

8 units → 152
1 unit → 152 ÷ 8 = 19

Step 3 : Find the number of sweets in each box

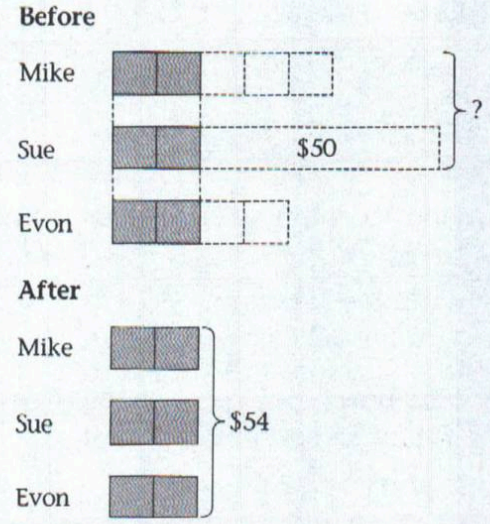
Box A → 6 × 19 = 114
Box B → 8 × 19 = 152

There were 114 sweets in box A and 152 sweets in box B at first.

Answers: Box A: 114 sweets
Box B: 152 sweets

Solution to Question 28

Step 1 : Draw a model



Step 2 : Find the amount of money for 1 unit

6 units → $54
1 unit → $54 ÷ 6 = $9

Step 3 : Find the total amount of money Mike and Sue had at first

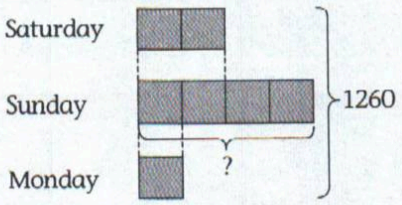
7 units → 7 × $9 = $63
$63 + $50 = $113

The total amount of money Mike and Sue had at first was $113.

Answer: $113

Solution to Question 29

Step 1 : Draw a model



Step 2 : Find the number of buns the baker sold on Monday

7 units → 1260
1 unit → 1260 ÷ 7
= 180

Step 3 : Find the number of buns the baker sold on Sunday

4 units → 4 × 180 = 720

He sold 720 buns on Sunday.

Answer: 720 buns

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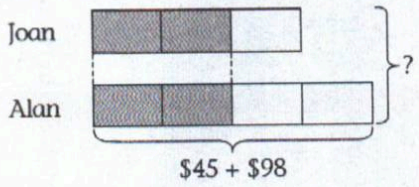
Solution to Question 30

Step 1 : Draw a model



Step 2 : Find the total amount of money Alan had

$45 + $98 = $143

Step 3 : Find the amount of money for 1 unit

4 units → $143
1 unit → $143 ÷ 4
= $35.75

Step 4 : Find the total amount of money Joan and Alan had

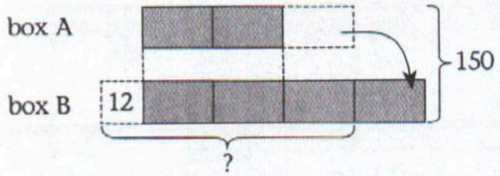
7 units → 7 × $35.75 = $250.25

Joan and Alan had $250.25 altogether.

Answer: $250.25

Solution to Question 31

Step 1 : Draw a model



Step 2 : Find the number of marbles for 1 unit

6 units → 150 – 12 = 138
1 unit → 138 ÷ 6 = 23

Step 3 : Find the number of marbles in box B in the beginning

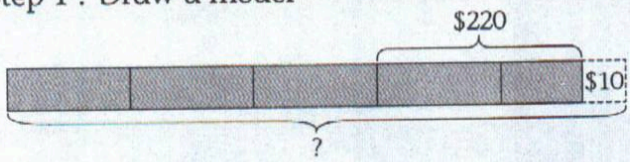
3 units → 3 × 23 = 69
69 + 12 = 81

There were 81 marbles in box B in the beginning.

Answer: 81 marbles

Solution to Question 32

Step 1 : Draw a model



Step 2 : Find the amount of money for 1 unit

2 units → $220 + $10 = $230
1 unit → $230 ÷ 2 = $115

Step 3 : Find the normal price of the watch

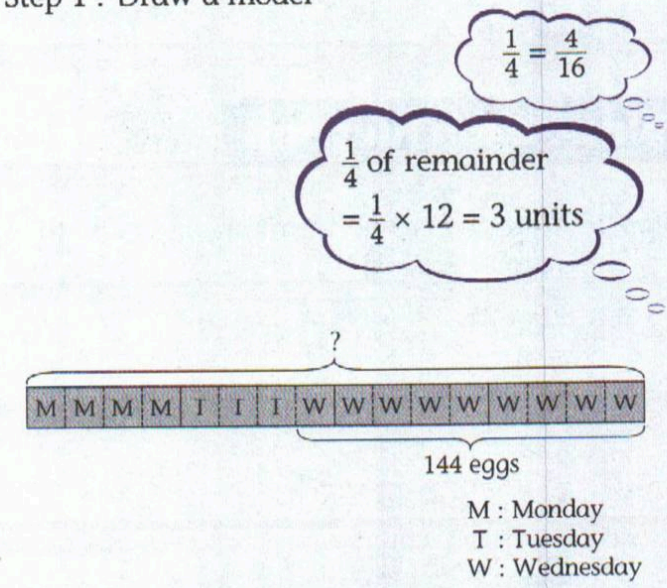
5 units → 5 × $115 = $575

The normal price of the watch was $575.

Answer: $575

Solution to Question 33

Step 1 : Draw a model



Step 2 : Find the number of eggs for 1 unit

9 units → 144
1 unit → 144 ÷ 9
= 16

Step 3 : Find the number of eggs he had in the beginning

16 units → 16 × 16 = 256

He had 256 eggs in the beginning.

Answer: 256 eggs

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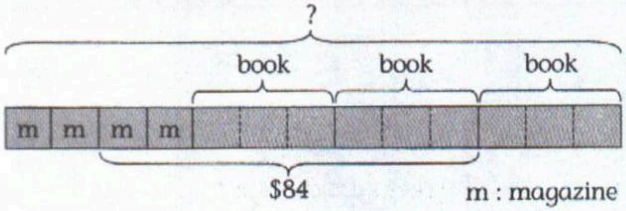
Solution to Question 34

Step 1 : Draw a model



Step 2 : Find the cost of one such magazine

8 units → $84.00

1 unit → $84.00 ÷ 8
= $10.50

Step 3 : Find the amount of money he had at first

13 units → 13 × $10.50 = $136.50

Sam had $136.50 at first.

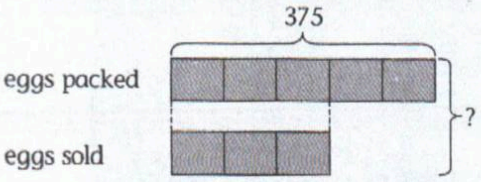
Answer: $136.50

Solution to Question 35

Step 1 : Find the total number of eggs the farmer packed

15 × 25 = 375

Step 2 : Draw a model



Step 3 : Find the number of eggs the farmer had at first

5 units → 375

1 unit → 375 ÷ 5 = 75

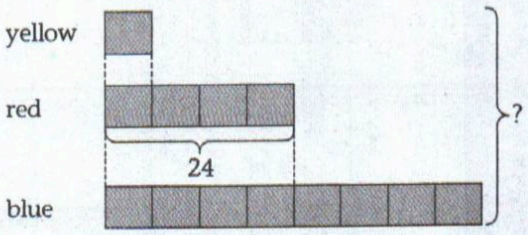
8 units → 8 × 75 = 600

The farmer had 600 eggs at first.

Answer: 600 eggs

Solution to Question 36

Step 1 : Draw a model



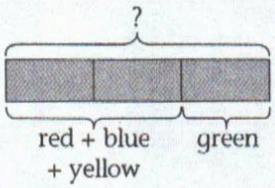
Step 2 : Find the total number of yellow, red and blue candles

4 units → 24

1 unit → 24 ÷ 4
= 6

13 units → 13 × 6
= 78

Step 3 : Find the total number of candles in the box



2 units → 78

1 unit → 78 ÷ 2
= 39

3 units → 3 × 39
= 117

There were 117 candles in the box altogether.

Answer: 117 candles

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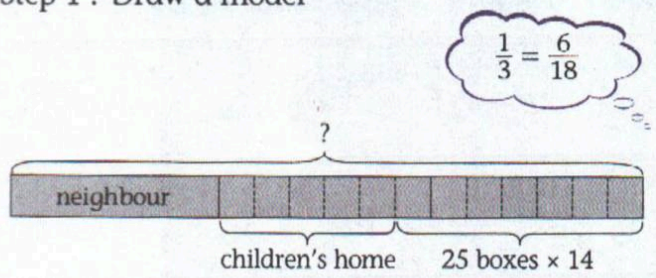
Solution to Question 40

Step 1 : Draw a model



Step 2 : Find the number of pies that she packed into boxes

$25 \times 14 = 350$

Step 3 : Find $\frac{1}{3}$ of Mrs Crawford's pies

7 units → 350
1 unit → $350 \div 7 = 50$
6 units → $6 \times 50 = 300$

Step 4 : Find the total number of pies Mrs Crawford baked

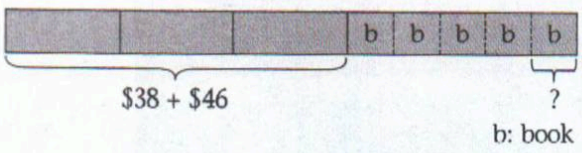
$3 \times 300 = 900$

Mrs Crawford baked 900 pies altogether.

Answer: 900 pies

Solution to Question 41

Step 1 : Draw a model



Step 2 : Find the amount of money Sandy spent on the bear and toy train

$\$38 + \$46 = \$84$

Step 3 : Find the amount of money she had left

3 units → \$84
1 unit → $\$84 \div 3$
= \$28
2 units → $2 \times \$28$
= \$56

Step 4 : Find the cost of each book

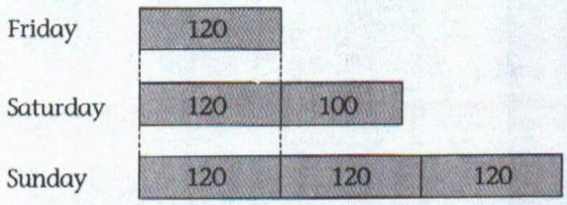
$\$56 \div 5 = \$11.20$

Each book cost \$11.20.

Answer: \$11.20

Solution to Question 42

Step 1 : Draw a model

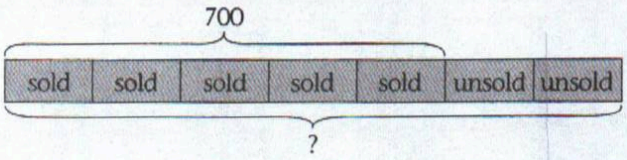


Step 2 : Find the total number of oranges he sold

$5 \times 120 = 600$

$600 + 100 = 700$

Step 3 : Draw another model



Step 4 : Find the number of oranges he had at first

5 units → 700
1 unit → $700 \div 5 = 140$
7 units → $7 \times 140 = 980$

He had 980 oranges at first.

Answer: 980 oranges

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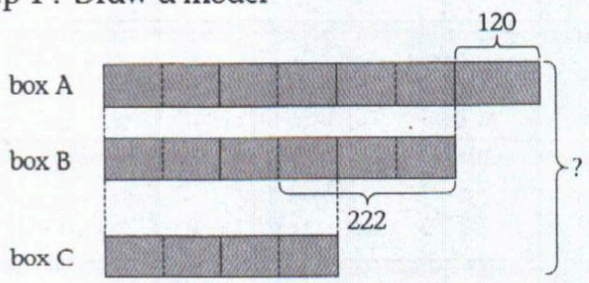
Solution to Question 43

Step 1 : Draw a model



Step 2 : Find the number of eggs for 1 unit

3 units → 222

1 unit → 222 ÷ 3 = 74

Step 3 : Find the total number of eggs

16 units → 16 × 74 = 1184

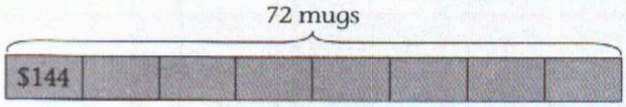
1184 + 120 = 1304

There were 1304 eggs altogether.

Answer: 1304 eggs

Solution to Question 44

Step 1 : Draw a model



Step 2 : Find the number of mugs Alfred sold

8 units → 72

1 unit → 72 ÷ 8 = 9

Step 3 : Find the cost of 1 mug

$144 ÷ 9 = $16

Step 4 : Find the amount of money he would get if he had sold all the mugs

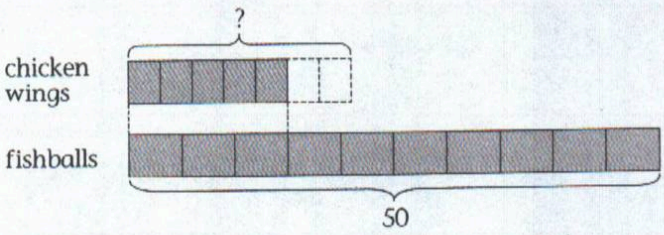
72 × $16 = $1152

He would get $1152 if he had sold all the mugs.

Answer: $1152

Solution to Question 45

Step 1 : Draw a model



Step 2 : Find the number of chicken wings left in the pack

10 big units → 50

1 big unit → 50 ÷ 10 = 5

3 big units → 3 × 5 = 15

Step 3 : Find the number of chicken wings in the pack at first

3 big units = 5 small units

5 small units → 15

1 small unit → 15 ÷ 5 = 3

7 small units → 7 × 3 = 21

There were 21 chicken wings in the pack at first.

Answer: 21 chicken wings

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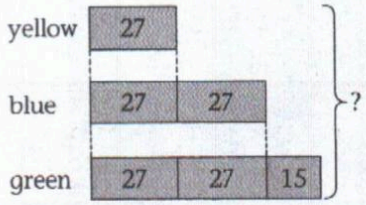
Solution to Question 46

Step 1 : Draw a model

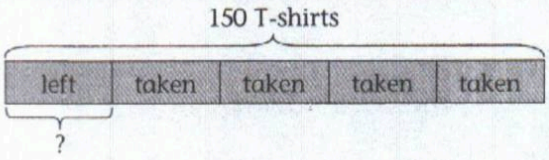


Step 2 : Find the total number of T-shirts

$5 \times 27 = 135$

$135 + 15 = 150$

Step 3 : Draw another model



Step 4 : Find the number of T-shirts left in the box

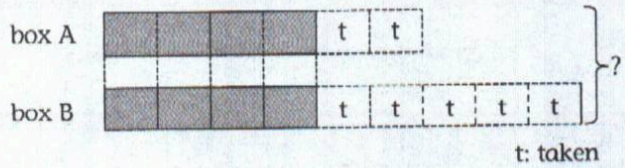
$150 \div 5 = 30$

30 T-shirts were left in the box.

Answer: 30 T-shirts

Solution to Question 47

Step 1 : Draw a model



Step 2 : Find the number of pears for 1 unit

7 units → 84

1 unit → $84 \div 7 = 12$

Step 3 : Find the number of pears in both boxes

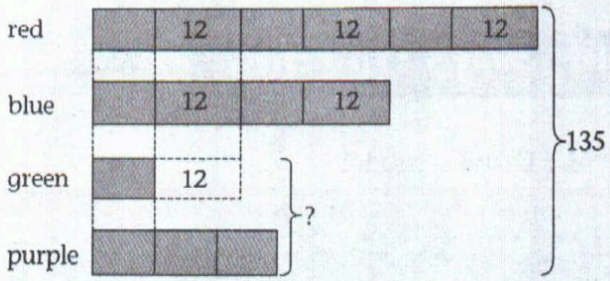
15 units → $15 \times 12 = 180$

There were 180 pears in both boxes at first.

Answer: 180 pears

Solution to Question 48

Step 1 : Draw a model



Step 2 : Find 1 unit of the purple balloons

$12 \times 6 = 72$

$135 - 72 = 63$

9 units → 63

1 unit → $63 \div 9 = 7$

Step 3 : Find the total number of purple and green balloons

4 units → $4 \times 7 = 28$

$28 + 12 = 40$

The total number of purple and green balloons Agnes had at first was 40.

Answer: 40 purple and green balloons

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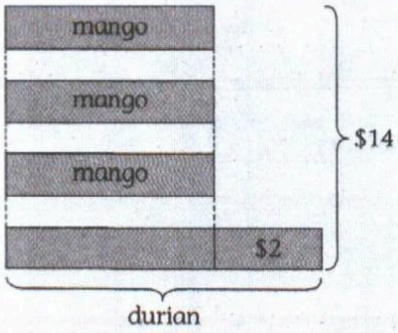
Solution to Question 49

Step 1 : Draw a model



Step 2 : Find the cost of 1 mango

$14 – $2 = $12

$12 ÷ 4 = $3

Step 3 : Find the cost of 1 durian

$3 + $2 = $5

Step 4 : Find the cost of 5 mangoes and 4 durians

5 mangoes → 5 × $3 = $15

4 durians → 4 × $5 = $20

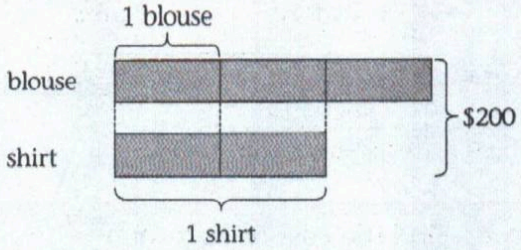
$15 + $20 = $35

The total cost of 5 mangoes and 4 durians was $35.

Answer: $35

Solution to Question 50

Step 1 : Draw a model

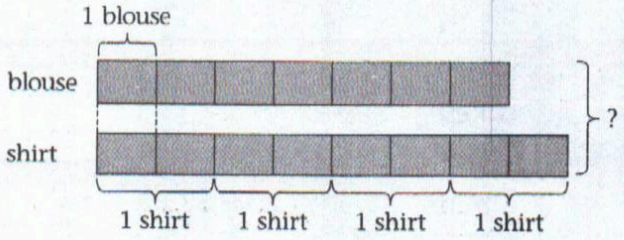


Step 2 : Find the cost of 1 blouse

5 units → $200

1 unit → $200 ÷ 5 = $40

Step 3 : Draw another model



Step 4 : Find the total cost of 7 blouses and 4 shirts

7 blouses + 4 shirts = 15 units

15 units → 15 × $40
= 3 × 5 × $40
= 3 × $200
= $600

The cost of 7 blouses and 4 shirts is $600.

Answer: $600

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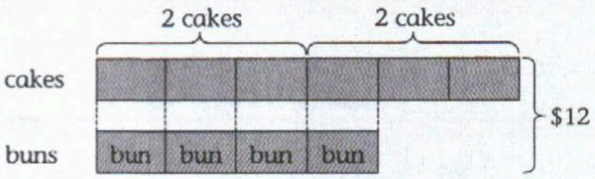
Solution to Question 51

Step 1 : Draw a model



Step 2 : Find the cost of each bun

10 units → $12

1 unit → $12 ÷ 10 = $1.20

Step 3 : Find the cost of each cake

3 × $1.20 = $3.60

$3.60 ÷ 2 = $1.80

Step 4 : Find the total cost of 6 buns and 9 cakes

6 buns → 6 × $1.20 = $7.20

9 cakes → 9 × $1.80 = $16.20

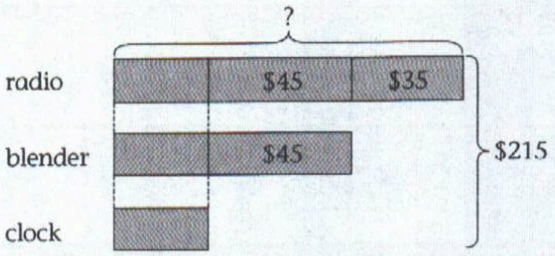
$7.20 + $16.20 = $23.40

The total cost of 6 buns and 9 cakes was $23.40.

Answer: $23.40

Solution to Question 52

Step 1 : Draw a model



Step 2 : Find the value of 3 units

$45 + $35 + $45 = $125

$215 – $125 = $90

Step 3 : Find the cost of the clock

3 units → $90

1 unit → $90 ÷ 3 = $30

Step 4 : Find the cost of the radio

$30 + $45 + $35 = $110

The cost of the radio is $110.

Answer: $110

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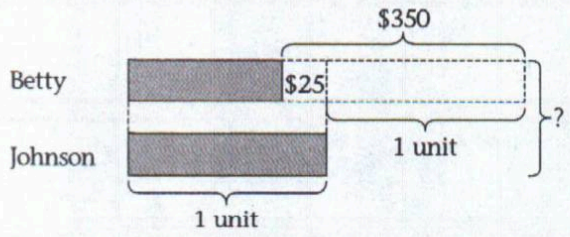
Solution to Question 53

Step 1 : Draw a model



Step 2 : Find the amount of money Johnson had

$350 – $25 = $325

Step 3 : Find the total amount of money they had at first

1 unit → $325

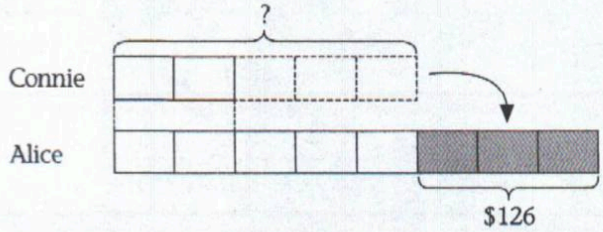
3 units → 3 × $325 = $975

They had $975 at first.

Answer: $975

Solution to Question 54

Step 1 : Draw a model



Step 2 : Find the amount of money for 1 unit

3 units → $126

1 unit → $126 ÷ 3 = $42

Step 3 : Find the amount of money Connie had at first

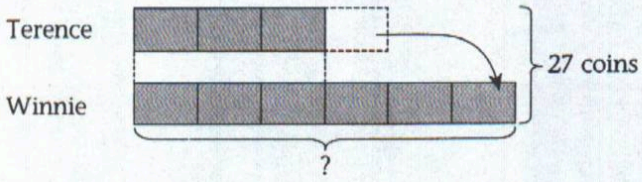
5 units → 5 × $42 = $210

Connie had $210 at first.

Answer: $210

Solution to Question 55

Step 1 : Draw a model



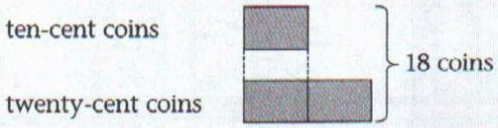
Step 2 : Find the number of coins Winnie had

9 units → 27

1 unit → 27 ÷ 9 = 3

6 units → 6 × 3 = 18

Step 3 : Draw another model

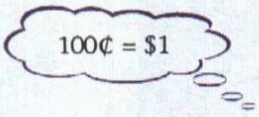


Step 4 : Find the number of ten-cent and twenty-cent coins Winnie had

ten-cent coins → 18 ÷ 3 = 6

twenty-cent coins → 2 × 6 = 12

Step 5 : Find the amount of money Winnie had in the end



6 × 10¢ = 60¢

12 × 20¢ = 240¢

60¢ + 240¢ = 300¢ = $3

She had $3 in the end.

Answer: $3

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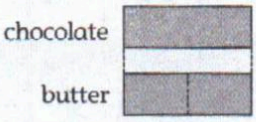
Solution to Question 56

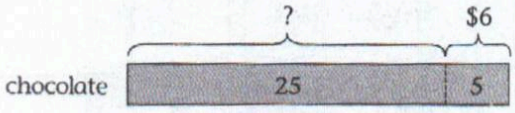
Step 1 : Draw a model to compare the prices of chocolate and butter cookies



20 butter cookies cost as much as 10 chocolate cookies.

Therefore, 20 butter cookies and 20 chocolate cookies cost as much as 30 chocolate cookies.

Step 2 : Draw a model to find the amount of money Hazel had at first



5 chocolate cookies → $6

1 chocolate cookie → 600¢ ÷ 5
= 120¢

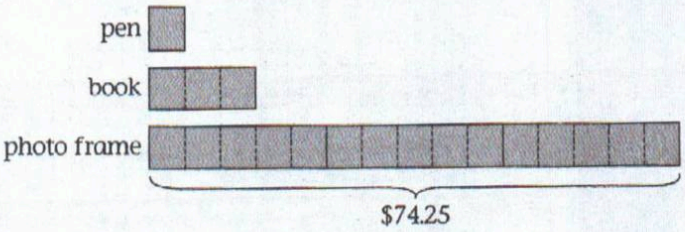
25 chocolate cookies → 25 × 120¢
= 3000¢
= $30

Hazel had $30 at first.

Answer: $30

Solution to Question 57

Step 1 : Draw a model



Step 2 : Find the cost of 1 unit

15 units → $74.25

1 unit → $74.25 ÷ 15 = $4.95

Step 3 : Find the cost of a book, a pen and a photo frame

9 units → 9 × $4.95 = $44.55

The total cost of a book, a pen and a photo frame is $44.55.

Answer: $44.55

Solution to Question 58

Step 1 : Find the amount of money he would receive if he sold 1 T-shirt, 10 T-shirts and 100 T-shirts respectively

1 T-shirt	$12
10 T-shirts	10 × $12 + $5 = $120 + $5 = $125
100 T-shirts	(100 × $12) + (10 × $5) + $50 = $1200 + $50 + $50 = $1300

Step 2 : Find the number of T-shirts he would have to sell if he wanted to receive $1372

$1372 – $1300 = $72

$72 ÷ $12 = 6

100 T-shirts + 6 T-shirts = 106 T-shirts

Joel would have to sell 106 T-shirts if he wanted to receive $1372 in total.

Answer: 106 T-shirts

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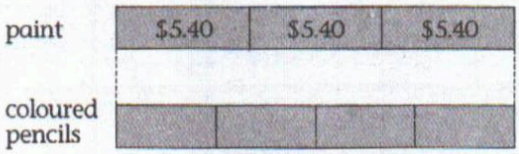
Solution to Question 59

Step 1 : Draw a model



Step 2 : Find the cost of 2 boxes of coloured pencils

$3 \times \$5.40 = \$16.20$

$\$16.20 \div 2 = \$8.10$

Step 3 : Find the amount of money Natalie had

$7 \times \$5.40 = \$37.80$

$\$37.80 + \$8.10 = \$45.90$

Step 4 : Find the amount of money Natalie would have left

$5 \times \$5.40 = \$27.00$

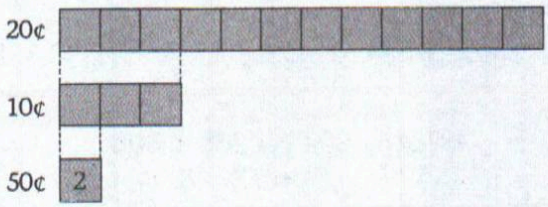
$\$45.90 - \$27.00 = \$18.90$

Natalie would have $18.90 left if she only bought 5 boxes of paint.

Answer: $18.90

Solution to Question 60

Step 1 : Draw a model



Step 2 : Find the number of coins in the different denominations

Number of ten-cent coins →
$3 \times 2 = 6$

Number of twenty-cent coins →
$12 \times 2 = 24$

Step 3 : Find the amount of money in the different denominations

Amount of money in fifty-cent coins →
$2 \times 50¢ = 100¢$

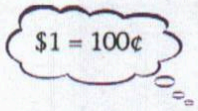
Amount of money in ten-cent coins →
$6 \times 10¢ = 60¢$

Amount of money in twenty-cent coins →
$24 \times 20¢ = 480¢$

Step 4 : Find the amount of money he had at first

$100¢ + 480¢ + 60¢ = 640¢$

Step 5 : Find the amount of money he had left if he spent $2.75 on a pair of socks



$640¢ - 275¢ = 365¢$
$= \$3.65$

He would have $3.65 left if he spent $2.75 on a pair of socks.

Answer: $3.65

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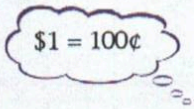
Solution to Question 61

Step 1 : Find the number of flowers Fanny had



$\$7.20 \div 80¢ = 720¢ \div 80¢$
$= 72 \div 8$
$= 9$

Step 2 : Find the amount collected for selling $\frac{2}{3}$ of the flowers at 90¢ each

$\frac{2}{\cancel{3}_{1}} \times \overset{3}{\cancel{9}} = 6$ flowers

$6 \times 90¢ = 540¢$

Step 3 : Find the amount collected for selling the rest of the flowers at 50¢ each

$9 - 6 = 3$ flowers

$3 \times 50¢ = 150¢$

Step 4 : Find the total amount of money Michael collected

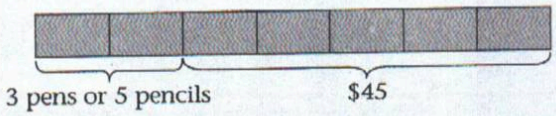
$540¢ + 150¢ = 690¢$
$= \$6.90$

Michael collected $6.90.

Answer: $6.90

Solution to Question 62

Step 1 : Draw a model



Step 2 : Find the cost of 3 pens or 5 pencils

5 units → $45

1 unit → $45 ÷ 5 = $9

2 units → 2 × $9 = $18

Step 3 : Find the cost of 1 pencil and 1 pen

pencil : $18 ÷ 5 = $3.60

pen : $18 ÷ 3 = $6

$3.60 + $6 = $9.60

The total cost of a pencil and a pen was $9.60.

Answer: $9.60

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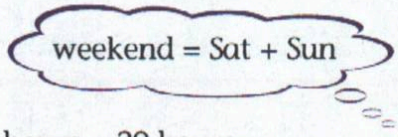
Solution to Question 63

Step 1 : Find the number of hours Ron worked for 10 weekdays

10 days × 8 hours = 80 hours

Step 2 : Find the number of hours Ron had worked for 4 days over 2 weekends



4 days × 5 hours = 20 hours

Step 3 : Find the amount of money Ron earned in 10 weekdays

80 hours × $12 = $960

Step 4 : Find the amount of money Ron earned in 4 days over 2 weekends

2 × $12 = $24

20 hours × $24 = $480

Step 5 : Find the amount of money Ron will earn in 2 weeks

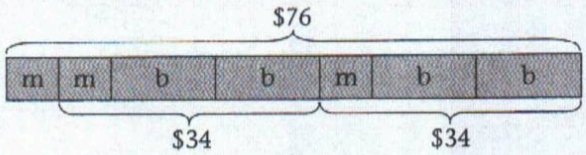
$960 + $480 = $1440

He will earn $1440 in 2 weeks.

Answer: $1440

Solution to Question 64

Step 1 : Draw a model



m : magazine
b : book

Step 2 : Find the cost of 1 magazine

$34 × 2 = $68

$76 – $68 = $8

Step 3 : Find the cost of 1 book

$34 – $8 = $26

$26 ÷ 2 = $13

Step 4 : Find the cost of 13 magazines and 9 books

13 magazines → 13 × $8 = $104

9 books → 9 × $13 = $117

Total → $104 + $117 = $221

She spends $221.

Answer: $221

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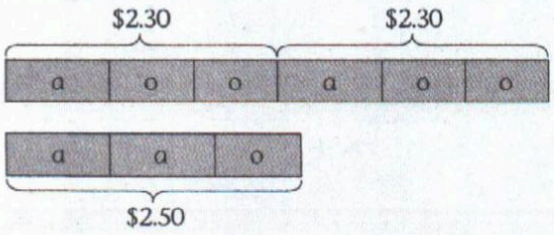
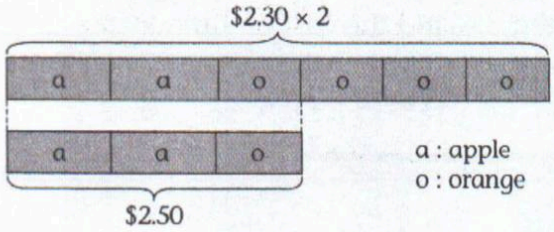
Solution to Question 65

Step 1 : Draw a model



Rearrange the model



Step 2 : Find the cost of 1 orange

$2.30 × 2 = $4.60

$4.60 – $2.50 = $2.10

$2.10 ÷ 3 = $0.70

Step 3 : Find the cost of 1 apple

$2.50 – $0.70 = $1.80

$1.80 ÷ 2 = $0.90

Step 4 : Find the cost of 4 apples and 6 oranges

4 × $0.90 = $3.60

6 × $0.70 = $4.20

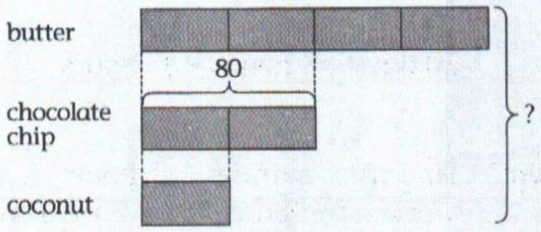
$3.60 + $4.20 = $7.80

The cost of 4 apples and 6 oranges is $7.80.

Answer: $7.80

Solution to Question 66

Step 1 : Draw a model



Step 2 : Find the total number of cookies

1 unit → 80 ÷ 2 = 40

7 units → 7 × 40 = 280

Step 3 : Find the number of boxes that she used to put the cookies in

280 ÷ 4 = 70

Step 4 : Find the total amount of money she collected

70 × $4.50 = 7 × 10 × $4.50
= 7 × $45
= $315

Mrs Bellamont would collect $315 from the sale of all the cookies.

Answer: $315

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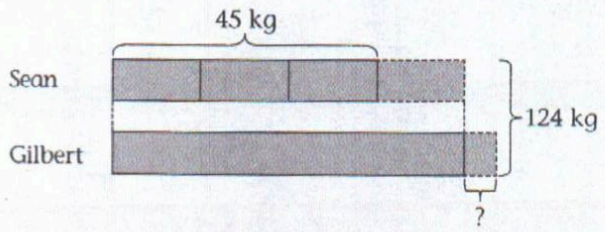
Solution to Question 67

Step 1 : Draw a model



Step 2 : Find the mass of Sean before he lost $\frac{1}{4}$ of his mass

3 units → 45 kg

1 unit → 45 kg ÷ 3 = 15 kg

4 units → 4 × 15 kg = 60 kg

Step 3 : Find the mass of Gilbert

124 kg – 60 kg = 64 kg

Step 4 : Find the difference in mass between Gibert and Sean

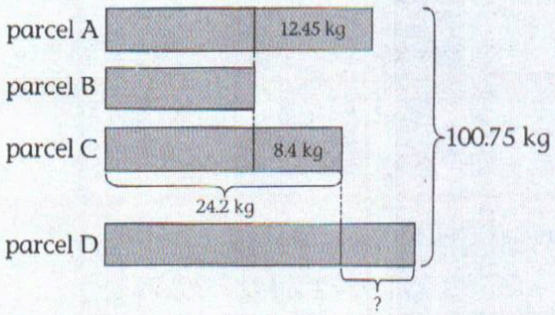
64 kg – 60 kg = 4 kg

Gibert was 4 kg heavier.

Answer: 4 kg

Solution to Question 68

Step 1 : Draw a model



Step 2 : Find the mass of parcel B

24.2 kg – 8.4 kg = 15.8 kg

Step 3 : Find the mass of parcel A

15.8 kg + 12.45 kg
= 28.25 kg

Step 4 : Find the total mass of parcels A, B and C

28.25 kg + 15.8 kg + 24.2 kg
= 68.25 kg

Step 5 : Find the mass of parcel D

100.75 kg – 68.25 kg
= 32.5 kg

Step 6 : Find how much heavier parcel D is than parcel C

32.5 kg – 24.2 kg
= 8.3 kg

Parcel D is 8.3 kg heavier than parcel C.

Answer: 8.3 kg

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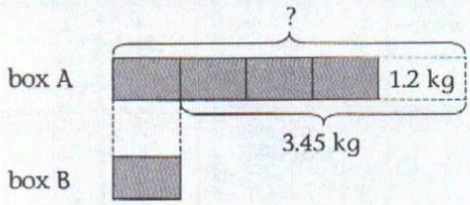
Solution to Question 69

Step 1 : Draw a model



Step 2 : Find the value of 1 unit

3.45 kg – 1.2 kg = 2.25 kg

3 units → 2.25 kg

1 unit → 2.25 kg ÷ 3 = 0.75 kg

Step 3 : Find the mass of box A

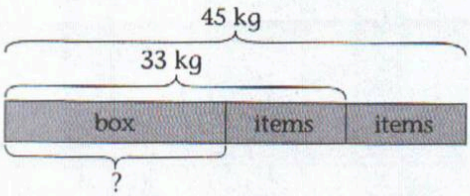
4 × 0.75 kg = 3 kg

3 kg + 1.2 kg = 4.2 kg

The mass of box A was 4.2 kg in the beginning.

Answer: 4.2 kg

Solution to Question 70



Step 2 : Find the mass of the remaining items

45 kg – 33 kg = 12 kg

Step 3 : Find the mass of the empty box

33 kg – 12 kg = 21 kg

Step 4 : Find the mass of 6 empty boxes

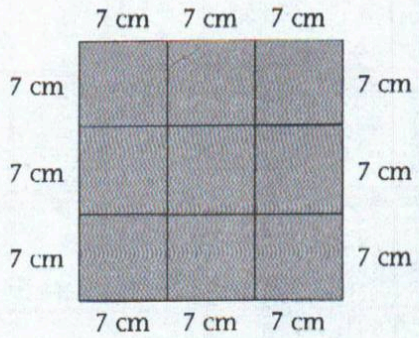
6 × 21 kg = 126 kg

The mass of 6 empty boxes is 126 kg.

Answer: 126 kg

Solution to Question 71

Step 1 : Draw the diagram of the large square



Step 2 : Find the length of each side of the large square

3 × 7 cm = 21 cm

Step 3 : Find the area of the large square

Area = Length × Breadth

21 cm × 21 cm = 441 cm^2

The area of the large square is 441 cm^2.

Answers: 441 cm^2

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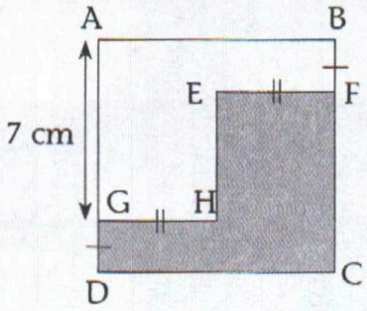
Solution to Question 72

Step 1 : Label the different lengths of the square



Step 2 : Find the length of each side

9 cm × 9 cm = 81 cm²

Length of AB = 9 cm

Length of BF = Length of GD
= 9 cm – 7 cm
= 2 cm

Length of EH = 7 cm – 2 cm = 5 cm

GH + EF = 9 cm

Step 3 : Find the perimeter of the unshaded portion

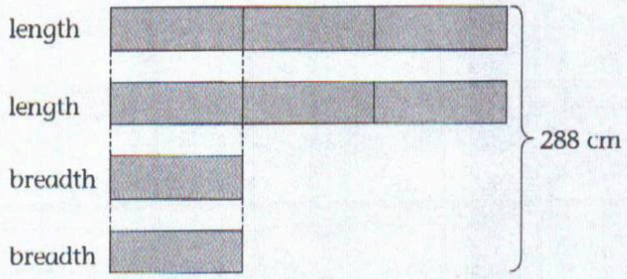
Perimeter of the unshaded portion
= AB + BF + (GH + EF) + EH + GA
= 9 cm + 2 cm + 9 cm + 5 cm + 7 cm
= 32 cm

The perimeter of the unshaded portion is 32 cm.

Answer: 32 cm

Solution to Question 73

Step 1 : Draw a model



Step 2 : Find the breadth of the rectangle

288 cm ÷ 8 = 36 cm

Step 3 : Find the length of the rectangle

3 × 36 cm = 108 cm

Step 4 : Find the area of the rectangle

36 cm × 108 cm = 3888 cm²

The area of the rectangle is 3888 cm².

Answer: 3888 cm²

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Solution to Question 74

Step 1 : Draw the diagrams of the square and rectangle

$\frac{2}{3}$ of the wire

$\frac{1}{3}$ of the wire

Step 2 : Find the length of each side of the square

$6 \text{ cm} \times 6 \text{ cm} = 36 \text{ cm}^2$

Step 3 : Find the length of the wire used to form a square

Length of wire used to make the square = Perimeter of the square

$4 \times 6 \text{ cm} = 24 \text{ cm}$

Step 4 : Find the total length of the wire

2 units → 24 cm

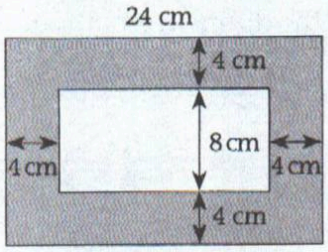
1 unit → 24 cm ÷ 2 = 12 cm

3 units → 3 × 12 cm = 36 cm

The wire was 36 cm long at first.

Answer: 36 cm

Solution to Question 75



Step 1 : Find the breadth of the paper

$8 \text{ cm} + 4 \text{ cm} + 4 \text{ cm} = 16 \text{ cm}$

Step 2 : Find the area of the paper

$24 \text{ cm} \times 16 \text{ cm} = 384 \text{ cm}^2$

Step 3 : Find the length of the picture

$24 \text{ cm} - 4 \text{ cm} - 4 \text{ cm} = 16 \text{ cm}$

Step 4 : Find the area of the picture

$16 \text{ cm} \times 8 \text{ cm} = 128 \text{ cm}^2$

Step 5 : Find the area of the border

$384 \text{ cm}^2 - 128 \text{ cm}^2 = 256 \text{ cm}^2$

The area of the border is 256 cm².

Answer: 256 cm²

Solution to Question 76

Step 1 : Find the perimeter of the rectangle

$15 \text{ cm} + 12 \text{ cm} = 27 \text{ cm}$

$2 \times 27 \text{ cm} = 54 \text{ cm}$

Step 2 : Find the area of the rectangle

$15 \text{ cm} \times 12 \text{ cm} = 180 \text{ cm}^2$

Step 3 : Find the perimeter of the square

$180 \text{ cm} \div 2 = 90 \text{ cm}$

Step 4 : Find the length of the wire needed to make 3 squares and 2 rectangles

3 squares → $3 \times 90 \text{ cm} = 270 \text{ cm}$

2 rectangles → $2 \times 54 \text{ cm} = 108 \text{ cm}$

Step 5 : Find the length of the wire

$270 \text{ cm} + 108 \text{ cm} + 20 \text{ cm} = 398 \text{ cm}$

The length of the wire is 398 cm.

Answer: 398 cm

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Solution to Question 77

Step 1 : Find the length of EH

EF = HG = 24 cm

EH and FG = 180 cm – 24 cm – 24 cm = 132 cm

EH = FG = 132 cm ÷ 2 = 66 cm

Step 2 : Find the length of rectangle ABCD

10 cm + 66 cm + 13 cm = 89 cm

Step 3 : Find the length of GZ

YH = 12 cm

GZ = 12 cm ÷ 2 = 6 cm

Step 4 : Find the breadth of rectangle ABCD

12 cm + 24 cm + 6 cm = 42 cm

Step 5 : Find the area of rectangle ABCD

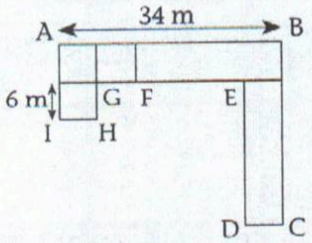
42 cm × 89 cm = 3738 cm^2

The area of rectangle ABCD is 3738 cm^2.

Answer: 3738 cm^2

Solution to Question 78

Step 1 : Label the diagram



Step 2 : Find the length of each rectangle

34 m – 6 m – 6 m = 22 m

Step 3 : Find the lengths of all sides of the garden

AB = 34 m
BC = 22 m + 6 m = 28 m
CD = 6 m
DE = 22 m
EF = 22 m – 6 m = 16 m
FG = 6 m
GH = 6 m
HI = 6 m
IA = 6 m + 6 m = 12 m

Step 4 : Find the perimeter of the garden

Perimeter of the garden =
34 m + 28 m + 6 m + 22 m + 16 m + 6 m
+ 6 m + 6 m + 12 m = 136 m

Step 5 : Find the cost of fencing the entire garden

2 m → \$23
1 m → \$23 ÷ 2 = \$11.50
136 m → 136 × \$11.50
= \$1564

Mr Cox will have to pay \$1564 to fence the entire garden.

Answer: \$1564

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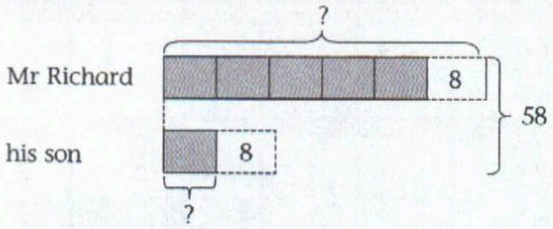
Solution to Question 79

Step 1 : Draw a model



Step 2 : Find their present total ages

$58 - 8 - 8 = 42$

Step 3 : Find his son's present age

6 units → 42

1 unit → $42 \div 6 = 7$

(a) His son's present age is 7 years old.

Step 4 : Find Mr Richard's age in 8 years' time

5 units → $5 \times 7 = 35$

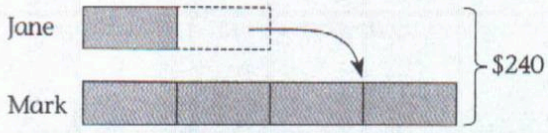
$35 + 8 = 43$

(b) Mr Richard will be 43 years old.

Answers: (a) 7 years old
(b) 43 years old

Solution to Question 80

Step 1 : Draw a model



Step 2 : Find the value of 1 unit

5 units → $240

1 unit → $240 ÷ 5 = $48

Step 3 : Find the amount of money Jane had at first

2 units → $2 \times \$48 = \$96$

(a) Jane had $96 at first.

Step 4 : Find the amount of money Mark had at first

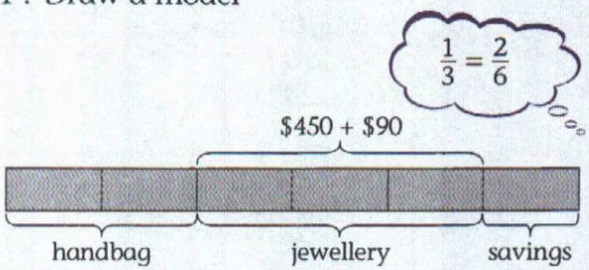
3 units → $3 \times \$48 = \$144$

(b) Mark had $144 at first.

Answers: (a) $96
(b) $144

Solution to Question 81

Step 1 : Draw a model



Step 2 : Find the amount of money she saved

$450 + $90 = $540

3 units → $540

1 unit → $540 ÷ 3 = $180

(a) She saved $180.

Step 3 : Find the amount of money she had at first

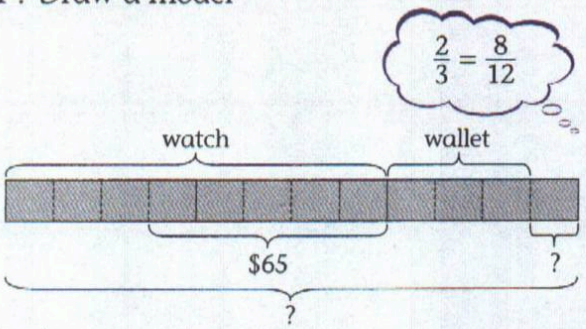
6 units → $6 \times \$180 = \$1080$

(b) She had $1080 at first.

Answers: (a) $180
(b) $1080

Solution to Question 82

Step 1 : Draw a model



Step 2 : Find the amount of money Sean had left

5 units → $65

1 unit → $65 ÷ 5 = $13

(a) Sean had $13 left.

Step 3 : Find the amount of money Sean had at first

12 units → $12 \times \$13 = \$156$

(b) Sean had $156 at first.

Answers: (a) $13
(b) $156

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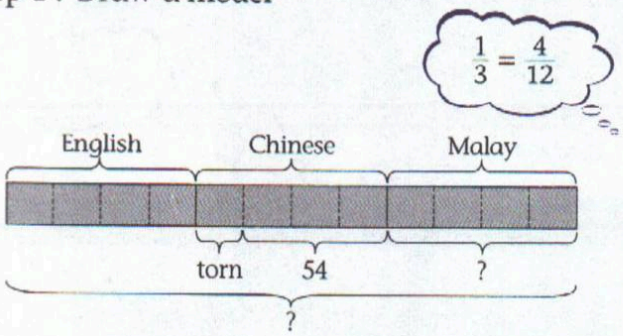
Solution to Question 83

Step 1 : Draw a model



Step 2 : Find the number of Malay books

3 units → 54

1 unit → 54 ÷ 3 = 18

4 units → 54 + 18 = 72

(a) There are 72 Malay books.

Step 3 : Find the total number of books

3 × 72 = 216

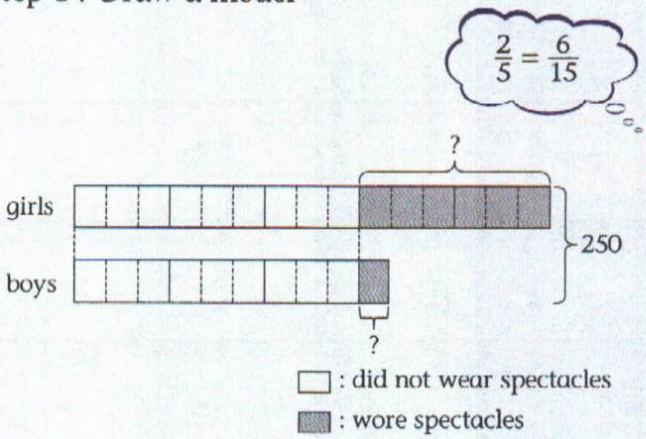
(b) There are 216 books altogether.

Answers: (a) 72 Malay books

(b) 216 books

Solution to Question 84

Step 1 : Draw a model



Step 2 : Find the number of boys who wore spectacles

25 units → 250

1 unit → 250 ÷ 25 = 10

(a) 10 boys wore spectacles.

Step 3 : Find the number of girls who wore spectacles

6 units → 6 × 10 = 60

(b) 60 girls wore spectacles.

Answers: (a) 10 boys

(b) 60 girls

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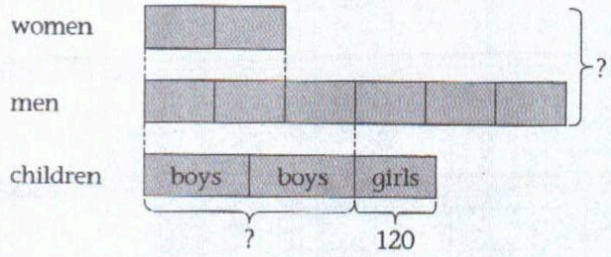
Solution to Question 85

Step 1 : Draw a model



Step 2 : Find the number of boys

$2 \times 120 = 240$

(a) There were 240 boys.

Step 3 : Find the number of adults

3 units → 240

1 unit → $240 \div 3 = 80$

8 units → $8 \times 80 = 640$

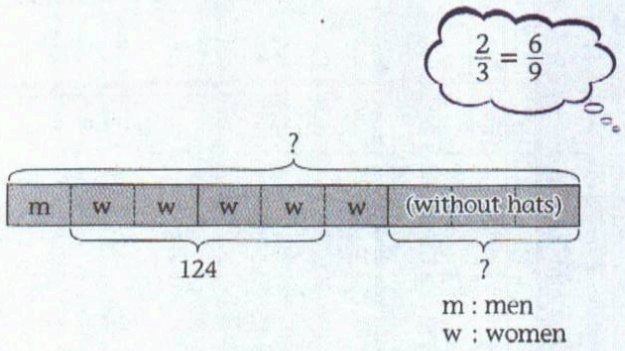
(b) There were 640 adults.

Answers: (a) 240 boys

(b) 640 adults

Solution to Question 86

Step 1 : Draw a model



Step 2 : Find the number of people who did not wear hats

4 units → 124

1 unit → $124 \div 4 = 31$

3 units → $3 \times 31 = 93$

(a) 93 people did not wear hats.

Step 3 : Find the total number of people at the fair

9 units → $9 \times 31 = 279$

(b) There were 279 people at the fair.

Answers: (a) 93 people

(b) 279 people

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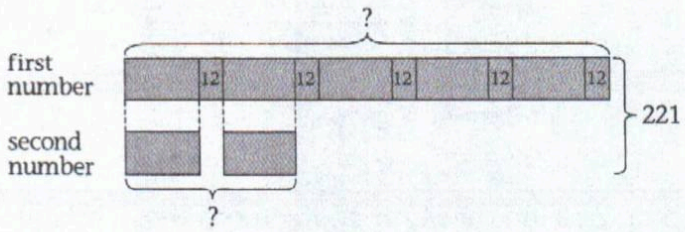
Solution to Question 87

Step 1 : Draw a model



Step 2 : Find the value of 1 unit

$5 \times 12 = 60$

$221 - 60 = 161$

7 units → 161

1 unit → $161 \div 7 = 23$

Step 3 : Find the first number

$23 + 12 = 35$

$5 \times 35 = 175$

(a) The first number is 175.

Step 4 : Find the second number

$2 \times 23 = 46$

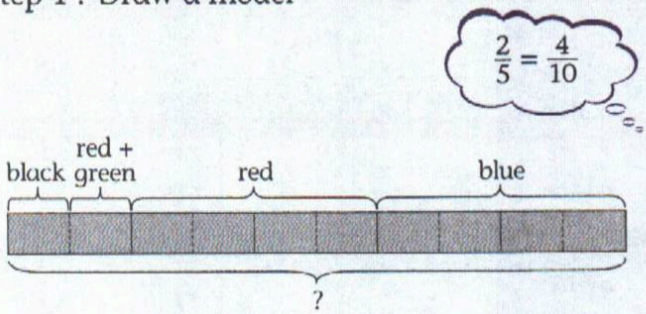
(b) The second number is 46.

Answers: (a) 175

(b) 46

Solution to Question 88

Step 1 : Draw a model



Step 2 : Find the total number of red and blue pens

1 unit → $24 + 12 = 36$

8 units → $8 \times 36 = 288$

$288 + 12 = 300$

(a) Gracia had 300 red and blue pens.

Step 3 : Find the total number of pens Gracia had

10 units → $10 \times 36 = 360$

(b) Gracia had 360 pens altogether.

Answers: (a) 300 red and blue pens

(b) 360 pens

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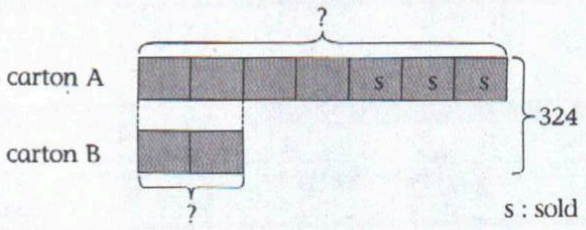
Solution to Question 89

Step 1 : Draw a model



Step 2 : Find the number of oranges in carton A at first

9 units → 324

1 unit → 324 ÷ 9 = 36

7 units → 7 × 36 = 252

(a) There were 252 oranges in carton A at first.

Step 3 : Find the number of oranges in carton B at first

2 units → 2 × 36 = 72

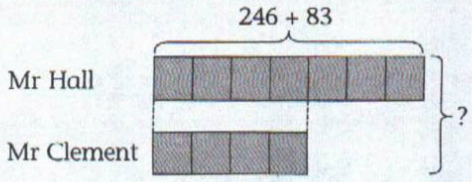
(b) There were 72 oranges in carton B at first.

Answers: (a) 252 oranges

(b) 72 oranges

Solution to Question 90

Step 1 : Draw a model



Step 2 : Find the number of lemons that Mr Hall sold on both days

246 + 83 = 329

Step 3 : Find the number of lemons Mr Clement sold on both days

7 units → 329

1 unit → 329 ÷ 7 = 47

4 units → 4 × 47 = 188

Step 4 : Find the number of lemons Mr Clement sold on Tuesday

Monday → 246 ÷ 2 = 123

Tuesday → 188 – 123 = 65

(a) Mr Clement sold 65 lemons on Tuesday.

Step 5 : Find the number of lemons both men sold altogether

329 + 188 = 517

(b) Both men sold 517 lemons altogether on both days.

Answers: (a) 65 lemons

(b) 517 lemons

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Solution to Question 91

Step 1 : Find the fraction of money that each of them saved every day

Cleo $\rightarrow 1 - \frac{4}{7} = \frac{3}{7}$

Diana $\rightarrow 1 - \frac{2}{3} = \frac{1}{3}$

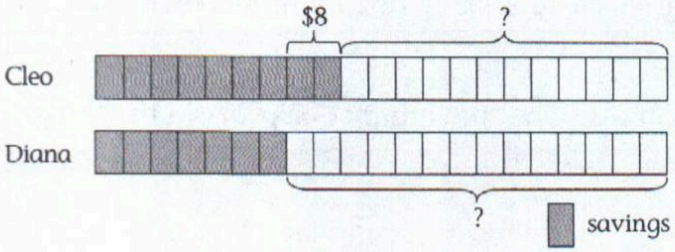
Step 2 : Find how much more Cleo saved than Diana in 1 day

$\$56 \div 7 = \$8$

Step 3 : Draw a model

Cleo $\rightarrow \frac{3}{7} = \frac{9}{21}$

Diana $\rightarrow \frac{1}{3} = \frac{7}{21}$



Step 4 : Find the amount of money each of them spent in a day

2 units $\rightarrow$ \$8

1 unit $\rightarrow$ $\$8 \div 2 = \$4$

12 units $\rightarrow$ $12 \times \$4 = \$48$

(a) Cleo spent \$48 a day.

14 units $\rightarrow$ $14 \times \$4 = \$56$

(b) Diana spent \$56 a day.

Answers: (a) \$48

(b) \$56

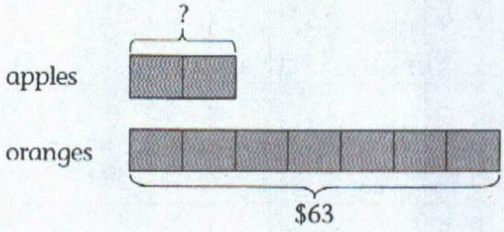
Solution to Question 92

Step 1 : Find the cost of 90 oranges

$90 \times 70¢ = 6300¢ = \$63$

\$1 = 100¢

Step 2 : Draw a model



Step 3 : Find the cost of a box of apples

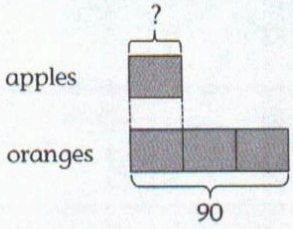
7 units $\rightarrow$ \$63

1 unit $\rightarrow$ $\$63 \div 7 = \$9$

2 units $\rightarrow$ $2 \times \$9 = \$18$

(a) A box of apples costs \$18.

Step 4 : Draw another model



Step 5 : Find the number of apples in 1 box

$90 \div 3 = 30$

(b) There are 30 apples in 1 box.

Answers: (a) \$18

(b) 30 apples

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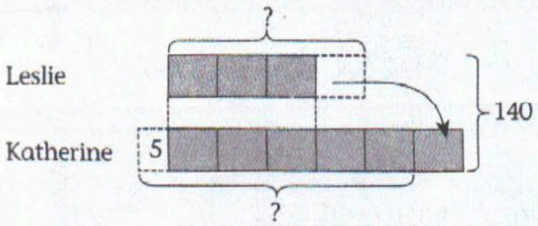
Solution to Question 93

Step 1 : Draw a model



Step 2 : Find the value of 1 unit

$140 - 5 = 135$

9 units → 135

1 unit → $135 \div 9 = 15$

Step 3 : Find the number of sweets Leslie had at first

4 units → $4 \times 15 = 60$

(a) Leslie had 60 sweets at first.

Step 4 : Find the number of sweets Katherine had at first

$140 - 60 = 80$

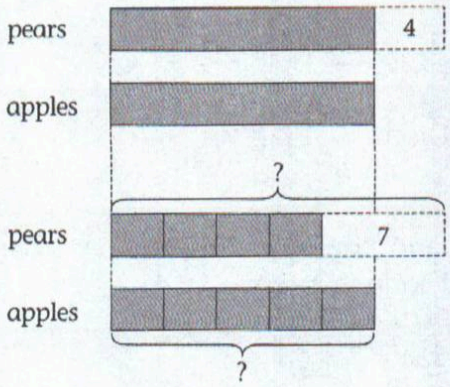
(b) Katherine had 80 sweets at first.

Answers: (a) 60 sweets

(b) 80 sweets

Solution to Question 94

Step 1 : Draw a model



Step 2 : Find the number of apples in the box at first

$7 - 4 = 3$

1 unit → 3

5 units → $5 \times 3 = 15$

(a) There were 15 apples in the box at first.

Step 3 : Find the number of pears in the box at first

4 units → $4 \times 3 = 12$

$12 + 7 = 19$

(b) There were 19 pears in the box at first.

Answers: (a) 15 apples

(b) 19 pears

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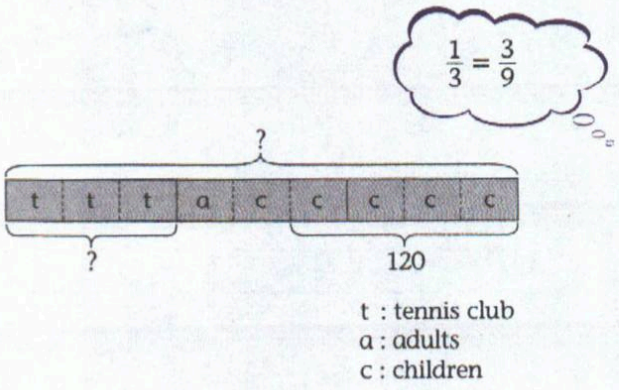
Solution to Question 95

Step 1 : Draw a model



t : tennis club
a : adults
c : children

Step 2 : Find the number of adults

4 units → 120

1 unit → 120 ÷ 4 = 30

Step 3 : Find the number of people belonging to the tennis club

3 units → 3 × 30 = 90

(a) 90 people belonged to the tennis club.

Step 4 : Find the total number of spectators

9 units → 9 × 30 = 270

(b) There were 270 spectators at the tennis match.

Answers: (a) 90 people

(b) 270 spectators

Solution to Question 96

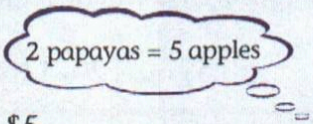
Step 1 : Find the cost of each papaya

18 – 12 = 6

$30 ÷ 6 = $5

(a) Each papaya cost $5.

Step 2 : Find the cost of three apples and three papayas



5 apples → 2 × $5
= $10

1 apple → $10 ÷ 5
= $2

3 apples → 3 × $2
= $6

3 papayas → 3 × $5
= $15

$6 + $15 = $21

(b) Three apples and three papayas would cost $21.

Answers: (a) $5

(b) $21

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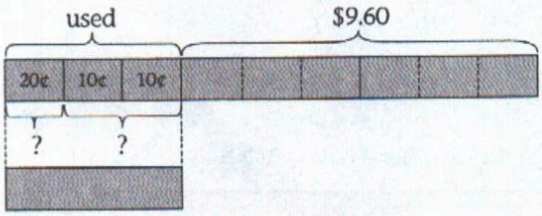
Solution to Question 97

Step 1 : Draw a model



Step 2 : Find the amount of money Joey used

$\$9.60 \div 2 = \$4.80$

Step 3 : Use 'Guess and Check' method to find how many ten-cent and twenty-cent coins she used

Number of 20¢ coins	Number of 10¢ coins	Total value of coins
4 $4 \times 20¢ = 80¢$	8 $8 \times 10¢ = 80¢$	$1.60
10 $10 \times 20¢ = 200¢$	20 $20 \times 10¢ = 200¢$	$4
12 $12 \times 20¢ = 240¢$	24 $24 \times 10¢ = 240¢$	**$4.80**

(a) She used 12 twenty-cent coins.

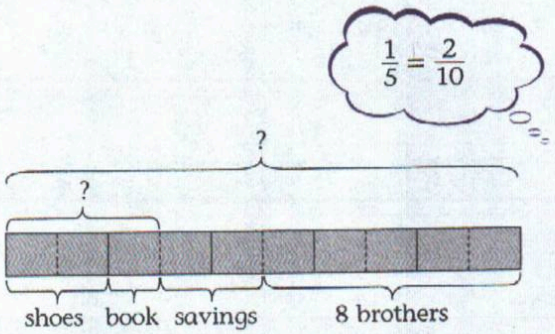
(b) She used 24 ten-cent coins.

Answers: (a) 12 twenty-cent coins

(b) 24 ten-cent coins

Solution to Question 98

Step 1 : Draw a model



Step 2 : Find the amount of money James had at first

$8 \times \$5 = \$40$

5 units → $40

1 unit → $\$40 \div 5 = \$8$

10 units → $10 \times \$8 = \$80$

(a) James had $80 at first.

Step 3 : Find the amount of money James spent

3 units → $3 \times \$8 = \$24$

(b) James spent $24.

Answers: (a) $80

(b) $24

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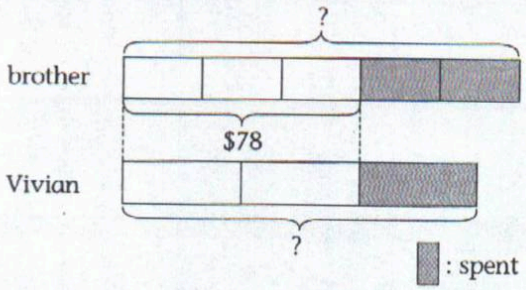
Solution to Question 99

Step 1 : Draw a model



Step 2 : Find the amount of money Vivian had at first

1 unit → \$78 ÷ 2 = \$39

3 units → 3 × \$39 = \$117

(a) Vivian had \$117 at first.

Step 3 : Find the amount of money Vivian's brother had at first

3 units → \$78

1 unit → \$78 ÷ 3 = \$26

5 units → 5 × \$26 = \$130

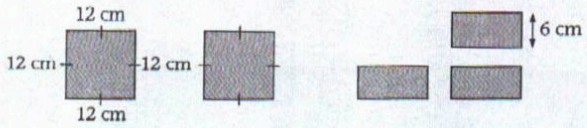
(b) Vivian's brother had \$130 at first.

Answers: (a) \$117

(b) \$130

Solution to Question 100

Step 1 : Draw the diagrams of the squares and rectangles



Step 2 : Find the length of the wire at first

4 × 12 cm = 48 cm

2 × 48 cm = 96 cm

2 × 96 cm = 192 cm

(a) The length of the wire is 192 cm.

Step 3 : Find the perimeter of each rectangle

96 cm ÷ 3 = 32 cm

Step 4 : Find the length of each rectangle

32 cm – 6 cm – 6 cm = 20 cm

20 cm ÷ 2 = 10 cm

Step 5 : Find the area of each rectangle

Area = Length × Breadth

= 10 cm × 6 cm

= 60 cm^2

(b) The area of each rectangle is 60 cm^2.

Answers: (a) 192 cm

(b) 60 cm^2

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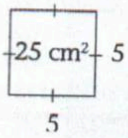
Solution to Question 101

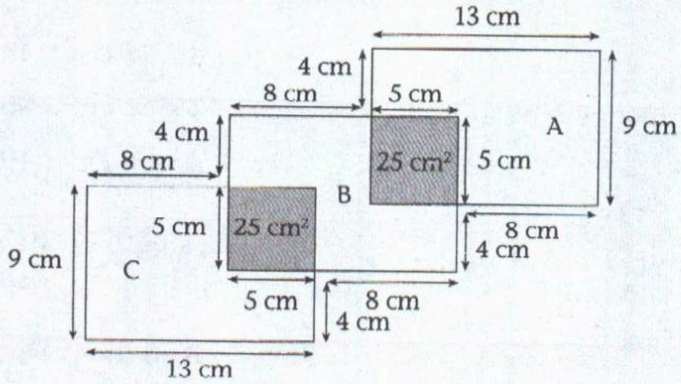
Step 1 : Find the length of each square



Area = Length × Breadth

5 cm × 5 cm = 25 cm²

Step 2 : Fill in the missing measurements and name the unshaded areas in the diagram



Step 3 : Find the perimeter of the whole figure

13 cm + 9 cm + 8 cm + 4 cm + 8 cm + 4 cm + 13 cm + 9 cm + 8 cm + 4 cm + 8 cm + 4 cm = 92 cm

(a) The perimeter of the whole figure is 92 cm.

Step 4 : Find the area of each rectangle

Area = Length × Breadth
= 13 cm × 9 cm
= 117 cm²

Unshaded area A = 117 cm² – 25 cm²
= 92 cm²

Unshaded area B = 117 cm² – 25 cm² – 25 cm²
= 67 cm²

Unshaded area C = 117 cm² – 25 cm²
= 92 cm²

Total unshaded area = 92 cm² + 67 cm² + 92 cm²
= 251 cm²

(b) The total area of the unshaded parts is 251 cm².

Answers: (a) 92 cm

(b) 251 cm²

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